Applied Energy 376 (2024) 124261

Contents lists available at ScienceDirect

Applied Energy

journal homepage: www.elsevier.com/locate/apenergy

A novel link prediction model for interval-valued crude oil prices based on complex network and multi-source information

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HIGHLIGHTS

- Predictive modeling to transform interval-valued into complex networks is proposed.
- A link prediction model based on multi-source information is constructed.
- Combining AI, link prediction, and complex networks forms a multidisciplinary mothed.
- The proposed model is well validated in predicting interval WTI crude oil prices.
- The constructed model can effectively extract multi-scale data features.

ARTICLE INFO

Keywords:
Interval-valued crude oil price
Directed visibility graph networks
Machine learning
Link prediction
Multi-source information

ABSTRACT

Accurate crude oil price forecasting is crucial for maintaining energy economy stability and enhancing investment and operational decision-making. Nowadays, the link prediction model for crude oil prices based on a complex network model has become an emerging and promising approach. However, existing link prediction methods are unable to extract complex information about interval-valued crude oil prices, and fail to consider the impact of multi-source information. Therefore, this paper proposes the link prediction model for interval-valued crude oil prices using complex network and multi-source information, which converts crude oil price interval series into complex networks. Through link prediction, it takes into account both the impact of past interval-crude oil prices on future interval time series and the interference of external real-time information on interval-valued prices. First, news headlines and geopolitical risk indices related to crude oil prices are captured, and search index data and news sentiment scores associated with interval data are analyzed. Subsequently, the interval sequence data and related influencing factors are decomposed and reconstructed respectively. The reconstructed interval center and radius sequence are then converted into a directed network, and node similarity in the network is calculated to generate similar nodes of the interval subsequence as feature values. Finally, multiple machine learning models are employed to acquire the optimal weighted combination prediction with interval prediction values. According to the experimental data, this approach outperforms other benchmark methods in terms of prediction accuracy.

1. Introduction

1.1. Research motivation

Crude oil, sometimes called the “blood of industry”, is an essential raw material for industrial development. Furthermore, crude oil plays a crucial role in the global economic system [1]. On a macroeconomic level, a spike in the price of crude oil can lead to higher inflation, more production costs, lower consumer purchasing power, and perhaps a recession. Numerous industries, including energy, aviation, transportation, and chemicals, are greatly affected by crude oil prices on an industry level. Instability in the supply chain, changes to investment decisions, and altered manufacturing costs are all possible outcomes of price swings. A dramatic increase in the price of crude oil and far-reaching effects on world politics and diplomacy could be the outcome of supply disruptions caused by geopolitical events.

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https://doi.org/10.1016/j.apenergy.2024.124261
Received 31 May 2024; Received in revised form 13 July 2024; Accepted 16 August 2024
Available online 27 August 2024

Furthermore, investor sentiment driven by economic indicators, geopolitical developments, and supply-demand forecasts leads to speculative trading, which in turn influences price volatility. When investors anticipate a future increase in oil prices, they may increase their investment in oil-related assets, thereby driving up oil prices. On the contrary, the expectations of future price declines may result in a decrease in investment. Therefore, investor expectations are closely related to oil price fluctuations, and accurately forecasting crude oil prices necessitates the ability to forecast investor behavior [2]. The complex and multi-faceted processes cause crude oil prices to be very volatile and nonlinear. Therefore, it is now a crucial yet intimidating undertaking to correctly forecast the price of crude oil.

Extensive research has been conducted on crude oil price forecasting due to its significance. A lack of investigation into interval-valued crude oil price forecasting exists in the research, which mostly concentrates on predicting the point closing prices. However, crude oil prices have daily, weekly, or monthly highs and lows and continuously fluctuate within a specific trading period, which leads to point closing price forecasting being inappropriate. Interval-valued crude oil price forecasting considers the fluctuation range between two consecutive time spots, thereby quantifying short-term fluctuations of crude oil prices [3–5]. Consequently, interval-valued crude oil price forecasting can consider variability and uncertainty information, offering policy-makers and market participants more decision-making information to make decisions [6].

The price of interval-valued crude oil exhibits cyclical fluctuations. However, current forecasting methods solely rely on up-to-date interval-valued crude oil price data, leading to a loss of historical information [7]. Scholars have focused on addressing the challenge of utilizing relevant historical interval data for prediction. Recently, the domain of time series forecasting has witnessed the emergence of an innovative methodology that converts time series into intricate networks for link prediction [8]. By calculating the similarity of nodes, this method can capture the correlation between historical and current data by identifying the most relevant past nodes to the current node [9]. Hence, we apply this method to the field of interval value prediction, with the aim of mitigating the disadvantage of ignoring historical interval data in existing interval value crude oil price prediction. In addition, future investigation is necessary to determine the feasibility of implementing in interval sequence forecasting.

1.2. Literature review

In recent years, scholars have proposed an increasing number of crude oil price forecasting methods. Those forecasting methods can be categorized into three types: single crude oil price forecasting methods, multi-source information crude oil price forecasting methods, and complex network link prediction methods. These methods each have their inherent advantages and disadvantages.

1.2.1. Single crude oil price forecasting methods

Traditional methods for predicting crude oil time series predominantly hinge on historical price data and fall into three main types: statistical prediction methods, machine learning prediction methods, and decomposition ensemble methods. Statistical models, noted for their wide applicability, operate under the assumption of linear relationships within time series data and frequently serve as benchmarks in comparison to more advanced predictive models. Typical statistical models include Vector Autoregression [10], Autoregressive Integral Moving Average [11], and Generalized Autoregressive Conditional Heteroscedasticity [12]. While these models demonstrate efficacy in capturing linear information within time series, they may struggle to capture nonlinear information [13]. To address those limitations and enhance the accuracy of crude oil price prediction, machine learning techniques are extensively employed, like back propagation neural networks (BPNN) [14], support vector machines (SVM) [15], and long short-term memory networks (LSTM) [16]. For instance, Chiroma et al. [17] proposed a West Texas Intermediate (WTI) crude oil prediction method based on a genetic algorithm and neural network. Simulation results indicate that the proposed method outperforms baseline algorithms in terms of prediction accuracy and computational efficiency. Despite that machine learning methods can capture non-linear relationships within time series, they also have disadvantages concerning parameter optimization and overfitting.

To solve those issues, decomposition ensemble methods have been widely applied in crude oil price prediction. Sun et al. [18] used an improved Complete Ensemble EMD with the Adaptive Noise (CEEMDAN) method to decompose and reconstruct the crude oil price sequence. The low-frequency and trend components were predicted using the chaotic sparrow search algorithm (CSSA) to optimize the kernel extreme value learning machine (KELM). The high-frequency components were further decomposed employing empirical mode decomposition (EMD) and then predicted with the CSSA-KELM model. The proposed method enhanced the accuracy of oil price prediction. Sen et al. [19] used particle swarm optimization to optimize the hyperparameters of LSTM and gated recurrent unit (GRU) for predicting crude oil prices. Moreover, Li et al. [20] proposed a novel hybrid method that integrates mean set empirical mode decomposition and mixed kernel extreme learning machine to model and predict crude oil prices, which significantly outperformed previous benchmark models in terms of prediction accuracy. After data was decomposed and rebuilt using variational mode decomposition, Zheng et al. [21] suggested decomposing and reconstructing data using variational mode decomposition, followed by predicting crude oil interval values using autoregressive conditional interval models and LSTM. However, these studies rely solely on historical data for prediction and ignore the impact of external factors on crude oil prices, such as policies, weather, and unexpected events, which leads to lag in fitting results. Therefore, this paper will consider combining decomposition ensemble with external factors to predict crude oil prices.

1.2.2. Crude oil price forecasting methods based on multi-source information

To improve the forecasting timeliness, several researchers have proposed multi-source information forecasting methods. By maximizing the complementarity and correlation between different data sources, this approach enhances the reliability and accuracy of prediction models. The most commonly used data sources in the field of crude oil price prediction originated from online and social media. Auxiliary variables such as the Internet search index [22] and online news [23] were employed in unstructured data studies to examine the real-time impact of qualitative factors. The Internet search index can timely reflect an individual's attention, behavior patterns, and psychological expectations, thus providing a lot of real-time and effective information for prediction. By integrating the Google Index and conventional economic data into their prediction models, Tang et al. [24] and Yang et al. [25] improved prediction accuracy and reflected the macro and micro mechanisms that influence crude oil prices, thereby reducing prediction bias. Extreme events such as political instability and economic growth can deeply impact the crude oil market. Text mining algorithms can capture market and social sentiment from online crude oil news, thereby extracting actionable information to enhance prediction accuracy [26–28]. To better forecast changes in crude oil prices across various news themes, Li et al. [29] proposed a feature grouping method based on the potential Latent Dirichlet Allocation (LDA) model to investigate the correlation between network media texts and crude oil price fluctuations. Hence, as the news media reflects significant events in the recent crude oil market, online search indices show the popularity of these events, and the fusion of these two types of data sheds even more light on market changes. Wu et al. [30] introduced an innovative data-driven approach for predicting crude oil prices, leveraging search indices and text mining of online media sources. The results indicate that the complementary relationship between news headlines and search indices helps to make fairly accurate crude oil price predictions. Furthermore, it is well known that geopolitics has a significant impact on crude oil prices, but it is difficult to measure using the above methods. Following Mei et al. [31] who applied the geopolitical risk (GPR) index to predict oil prices and verify the strong correlation between the two, this paper uses structured data GPR as an indicator to measure geopolitical events. However, in summary, although there have been some breakthroughs in academic research on using multiple sources of information to predict crude oil prices, the majority of literature generally focuses on predicting the closing price of point values rather than interval values. By displaying both the highest and lowest potential values of a target, interval prediction effectively conveys the inherent uncertainty and fluctuation range in predictions. Hence, for dynamic and complex crude oil price data, accurately forecasting the range of price fluctuations can offer valuable insights to managers. This study will make predictions based on interval crude oil prices.

1.2.3. *Complex network link prediction methods*

Researchers have paid a lot of attention to new methods for complex network link forecasting methods. Many studies have demonstrated that complex networks can be employed to mine effective information that traditional time series prediction cannot access, thus improving the predictive performance of time series [32]. Recently, the method of converting time series data into complex networks and using network analysis techniques to predict time series has been applied in various fields such as crude oil price prediction [33] and carbon price prediction [34,35]. By constructing complex network structures and utilizing link prediction techniques, it is possible to capture nonlinear and long-term dependencies in time series. Empirical studies have shown that the traditional method of using finite lag data as prediction eigenvalues only considers the impact of recent data on the future while ignoring past effective information [7]. However, time series such as crude oil prices and carbon prices exhibit periodicity and nonlinearity [36]. The patterns of past data changes are likely to repeat in the future. This method can effectively capture this similarity relationship by mapping data into a network and leveraging the node similarity in the network. In addition, this method can make use of the topology and similarity indicators in the network to more comprehensively and flexibly capture nonlinear relationships [37], thereby increasing prediction accuracy.

The process of predicting and connecting the next unknown node in a network using similar nodes as feature values is called link prediction. The goal of link prediction is to predict new edges that may appear in the future or restore missing edges in the existing graph based on the existing network structure [38]. Converting time series into networks and utilizing network analysis techniques for prediction can be seen as an application of link prediction [39]. Thus, converting a time series into a network for prediction involves two main stages: converting the time series into a network and calculating the similarity of network nodes. First, time series values are connected as nodes on the basis of specific connection conditions, and the connected network retains both the characteristics of the time series and its fluctuation properties. The visibility algorithm proposed by Lacasa et al. [40] is considered an important creation in this field, which maintains low computational complexity when converting time series into networks with nodes and edges, and has advantages in practical applications [41]. Secondly, by using the calculation method of node similarity in link prediction to calculate the similarity between nodes in the network, the relationship between past and nearest nodes is explored for predicting future nodes. There are numerous methods for calculating node similarity, which are broadly classified as path dependency methods (e.g., random walk) and common node dependency methods (e.g., common neighbors and Adam/Adar). Studies have demonstrated that path dependent methods outperform node dependent methods in link prediction [42]. The local random walk algorithm performs the best among them [43] Mao et al. [7] took a fresh look at visibility graph networks by using visibility algorithms to convert data into nodes, calculating node similarity through a local random walk algorithm, and establishing a prediction model based on the mutual visibility and similarity of the constructed carbon network. Recently, Liang et al. [44] proposed a hybrid framework that combines artificial intelligence, link prediction, and complex networks for predicting port container throughput, significantly improving prediction accuracy. By integrating AI with complex networks, Ren et al. [45] analyzed chaotic time series and converted them into graph signals for prediction using phase space embedding to deliver satisfactory results. In traditional models, employing only finite lag data as eigenvalues might lead to the loss of past information. Hence, complex network link prediction benefits from taking into account the relationship between previous and current information. These before-mentioned studies transform time-series data into complex networks and employ link prediction methods to extract valuable information from previous time points, thereby enhancing prediction effectiveness. While this method has made significant progress in multiple prediction fields, it is only applicable to predicting the closing price of crude oil prices, and interval link prediction methods need to be explored. The existing prediction methods of complex network link prediction rely solely on historical data and do not account for the real-time impact of exogenous variables, thus this paper will further improve the accuracy of predictions from this perspective.

1.3. *Interval-valued crude oil price forecasting*

Interval-valued time series have excellent advantages in coping with system uncertainty and they have received a lot of attention in academia. Interval prediction provides a range of possible future values for a variable, allowing decision-makers to more accurately reflect the uncertainty of the prediction and better manage risks. A lot of literature indicates that it is widely used in the prediction and analysis of stock prices [46], photovoltaics [47], wind power [48–50], and other related fields. It is also used in predicting crude oil prices. More specially, Sun et al. [51] proposed an interval decomposition ensemble learning method that combines binary empirical mode decomposition, interval multilayer perceptron, and interval exponential smoothing to predict interval crude oil prices; Wang et al. [52] established a memory-based hybrid prediction system for crude oil price range prediction; Sun et al. [53] proposed a crude oil price interval prediction method based on three-way clustering and decomposition ensemble learning. Data processing for interval-valued time series typically involves two methods: Minmax and CRM. The former divides the interval series into upper and lower bounds, while the latter divides the interval series into centers and radii. Wang et al. [54] compared these two different interval value data processing methods and found that using CRM achieved better prediction results. Therefore, this paper separated the interval values into centers and radii for prediction analysis. To sum up, significant progress has been made in predicting interval-valued crude oil prices in existing research. However, the existing interval prediction does not take into account the influence of past data and only uses recent limited lag data as input for prediction feature values. As a result, by combining link prediction of complex networks with interval crude oil price prediction, the fluctuation information of interval sequence values can be further explored in the network topology structure.

1.4. *Research gap*

Some crude oil price predictions have yielded positive results based on a review of relevant work, but due to the highly nonlinear and unstable characteristics of crude oil price data, this remains a challenging issue that merits further investigation. This section summarizes some limitations and research gaps in predicting crude oil prices, as follows:

(1) Most of the existing crude oil prices are based on point prediction, with limited research on interval prediction. At the same time, existing interval prediction only considers lagged interval data as

input for prediction eigenvalues, ignoring past effective interval information.

(2) Current research on converting time series into complex networks for link prediction only considers the influence of historical data on prediction, while the influence of external multi-source information is ignored.

(3) In the field of prediction, the existing prediction techniques which transform time series into network links have been proven to be feasible and effective. However, the research on combining interval crude oil price prediction with complex network link prediction is still in its early stages and deserves further exploration.

To bridge the research gaps mentioned above, a four-part proposal for interval-valued crude oil price link prediction is put forward based on complex network and multi-source information. The first step is to collect data from multiple sources like the Internet search index, news headlines, and geopolitical risk index. Then, the news headlines are quantified into sentiment scores by the Snownlp. Furthermore, the correlation between the multi-source information and interval-valued crude oil prices is examined to identify critical influencing factors. Step two involves transforming the interval crude oil price series into the center-radius format. To reduce the data complexity, the central sequence, radius sequence, and related influencing factors are individually decomposed into finite more stationary subsequences. In the third part, to characterize the complex fluctuations of crude oil prices, the directed visibility graph networks (hereinafter referred to as "DVGnets") are employed to respectively transform the decomposed center subsequence and radius subsequence into complex network form. After that, we identify how similar each node is in the network and use the ones that are most similar as features to feed into the forecasting model. Finally, to learn the linear and nonlinear correlations of the interval-valued crude oil prices series from multiple perspectives, four different forecasting methods are used to predict the subsequences. Afterward, the ideal interval prediction value is obtained by the optimal weighted combination method.

1.5. Main contributions

The main contributions of this paper are summarized as follows.

(1) A novel link prediction for interval-valued crude oil price based on complex network and multi-source information is proposed, which achieves higher prediction accuracy by incorporating multi-source information. Furthermore, by transforming interval-valued crude oil prices into complex network form, this method can effectively extract inherently complex data characteristics.

(2) A brand-new link prediction model for interval-valued crude oil price is constructed, which exhibits superior capability in capturing complex interval information of interval variables. Compared to existing crude oil price prediction models, the interval data used by the proposed method is transformed into nodes in the network, and past and recent data information is extracted by calculating the similarity of nodes, rather than using finite lag data as input, this novel approach avoids the loss of long-term historical information and allows for more precise prediction of future events or trends within specific time periods.

(3) A link prediction model based on multi-source information is developed, which effectively characterizes real-time fluctuations of interval-valued crude oil prices. To the best of our knowledge, this is the initial investigation to apply link prediction modeling in processing multi-source information. Implementing this approach can significantly enhance the informational capacity of the intricate network, enabling precise characterization of the instantaneous variations in interval-valued crude oil prices.

1.6. Organization and structure

The subsequent sections of this paper are organized as follows. Section 2 introduces the theoretical framework and related methods of combination prediction models. Section 3 provides an example of international crude oil price forecasting. Section 4 describes a detailed introduction to the numerical comparison experiment results and discussion. Section 5 presents the conclusions of the research.

2. Link prediction model for interval-valued crude oil price

Commencing this segment, we explicate the fundamental notion underlying international crude oil price intervals. Building upon this foundation, we introduce the link prediction model for interval-valued crude oil prices based on complex network and multi-source information while the specific models and procedures are outlined in detail. Finally, we discuss the feasibility of our approach.

2.1. Interval-valued time series

Assume the daily interval-valued international crude oil price $\widetilde{x} = [x_l, x_u] = \{x | x_l \leq x \leq x_u\}, x_l, x_u \in R$ where x_l and x_u are the daily low and high prices of crude oil. For interval-valued data $\widetilde{x}$, the interval radius $r_{\widetilde{x}}$ refers to the daily variation range of international crude oil prices, and the Interval center x_m denotes the average of the upper and lower limits of crude oil prices in the interval as shown in Fig. 1.

$$\begin{cases} r_{\widetilde{x}} = (x_u - x_l)/2 \\ x_m = (x_u + x_l)/2 \end{cases} \tag{1}$$

Based on the daily interval-valued international crude oil price, let $\widetilde{x} = [x_l(t), x_u(t)], t = 1, 2, \cdots, n$, and $\widetilde{x}(t)$ is the interval-valued time series of the international crude oil price. Fig. 2 shows the daily fluctuation range of WTI crude oil prices from November 1, 2023, to November 7, 2023. There is a wide range of crude oil prices, with large swings and extreme volatility between the two extremes. Therefore, compared to conventional point-value time series, daily interval-value time series offer a more comprehensive depiction of price levels and market dynamics, mitigating the loss of uncertain information.

2.2. Modeling framework

It is more challenging to make accurate predictions due to the high complexity and randomness of the international crude oil price time series. Consequently, the link prediction model for interval-valued crude

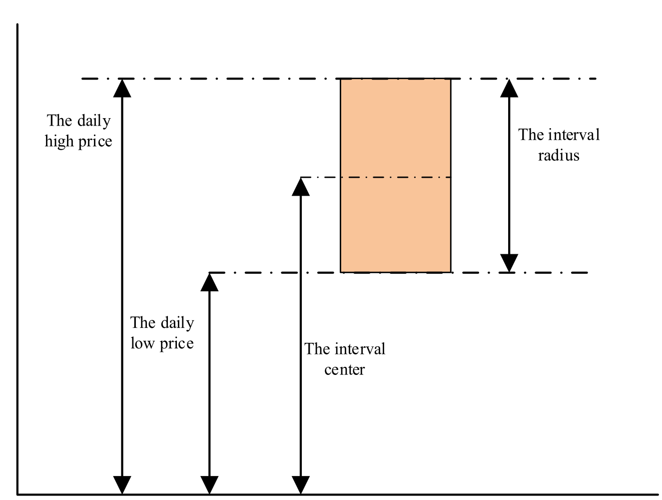

Fig. 1. Interval structure.

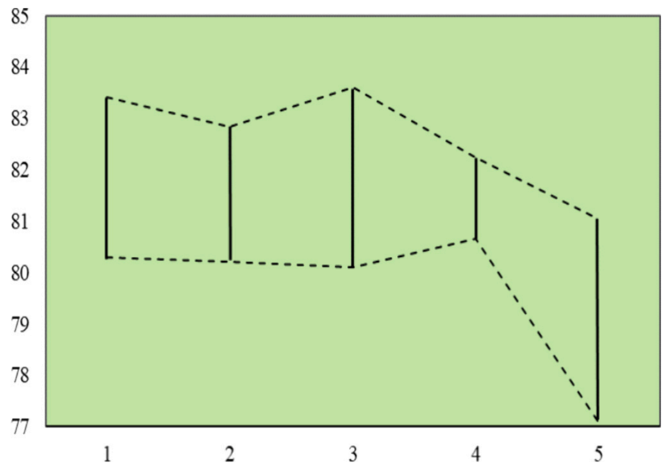

Fig. 2. Corresponding daily crude oil price intervals.

oil prices based on complex network and multi-source information is proposed. Fig. 3 shows the five portions that make up the model framework's specific procedure.

Step 1: Unstructured data processing. To begin, gather news headlines from websites connected to the international crude oil price. Subsequently, preprocess these headlines, and use Snownlp to perform sentiment analysis to derive text sentiment values. Simultaneously, based on the LDA, news text themes are divided to generate candidate keywords related to international crude oil prices, which are then crawled employing Baidu Index data. After calculating the Spearman correlation coefficients between the center/radius of oil prices and the candidate keywords and determining the final relevant keywords, the Principal Component Analysis (PCA) is utilized for dimensionality reduction. Finally, unstructured data about the range of crude oil prices can be collected.

Step 2: Structured data processing. Geopolitical factors have a substantial impact on international crude oil prices as a result of the volatile international environment. As a result, by crawling relevant websites and calculating the Spearman correlation coefficient between the GPR index and the interval- value of crude oil prices, it is possible to ascertain that the two are significantly correlated and to employ the GPR index as a structured influencing factor about the interval-value of crude oil prices.

Step 3: Interval multiscale decomposition. First, the data on the central price and its related influencing factors are decomposed using the Hilbert Huang transform (HHT); Second, the CEEMDAN approach to breaking out the radius pricing and the elements that affect it; Finally, as the last step in simplifying the sequence, the Sample Entropy (SE) is applied to reassemble the decomposed sequence into corresponding high-frequency, low-frequency, and trend terms.

Step 4: Calculation of similar nodes in a directed network. First, the center and radius values of crude oil prices at each frequency after decomposition are plotted using DVGnets to create directed networks; Next, the local random walk algorithm is applied to determine how similar each node is to the terminal nodes in the network. Finally, the first few nodes that are most similar to the terminal nodes are selected as feature values.

Step 5: Interval combination prediction. First, utilize the BPNN, Support Vector Regression (SVR), GRU, and LSTM to extract multi-scale data features for each subsequence, and the high-frequency, low-frequency, and trend item sequences of the center and radius are predicted one by one. Secondly, calculate the sum of the predicted values of the subsequences to obtain the center and radius values predicted by each model. The above combination prediction can extract scale aspects of interval-valued crude oil price time series from different perspectives using varied methods. Therefore, the prediction results corresponding to various prediction methods have complementary effects [55]. Ultimately, the optimal weight is computed through the optimal weighted combination technique to derive the optimal interval prediction result [6].

2.3. Feature processing

Geopolitical events exert a substantial influence on international crude oil prices. The GRP index, as a quantitative indicator, evaluates the impact of global political dynamics on the economy and markets. Using this index as structured data facilitates a more profound comprehension of how geopolitical factors affect fluctuations in crude oil prices. In addition, in the context of the Internet, historical data and economic indicators have a time lag, which makes the identification of the dynamic changes in international crude oil prices fuzzy. Unstructured data is useful for evaluating qualitative elements and provides immediate insights into investor attitudes and emotional changes related to occurrences concerning worldwide crude oil price data. Hence, our approach involves processing unstructured data from two perspectives: filtering search keywords and conducting sentiment analysis on news headlines.

2.3.1. Search keyword screening

Selecting suitable search terms is pivotal for capturing unbiased and comprehensive features of interest, which directly affects the accuracy of predictions. Hence, we propose a keyword filtering approach that relies on news headlines to accurately reflect the most relevant search queries on global crude oil prices. We utilize LDA to uncover potential thematic information within news title text datasets. LDA, as an unsupervised machine learning technique proposed by Blei et al., is widely used in domains like text mining and natural language processing to discover hidden topic structures in text sets [56]. It can reveal potential thematic information in text datasets, helping us identify key factors affecting crude oil prices. Specifically, the LDA model infers the themes and their related word distributions contained in each document by analyzing the co-occurrence frequency of words in the document, which is particularly effective in capturing investor hotspots and high-frequency search terms. Compared to other methods such as TF-IDF or N-gram models, LDA can capture not only frequency information of words but also identify potential thematic relationships of words, providing deeper text understanding and analysis [57]. Therefore, it has significant advantages in text mining and natural language processing and can produce good results in using LDA for data processing in this study.

The specific process of this section is as follows: Initially, we analyze the factors influencing crude oil prices and identify the preliminary keywords. Next, leveraging news headlines, we employ the LDA approach to extract candidate keywords while taking shifts into account in investor attention towards trending topics to acquire real-time comprehensive attention data. Subsequently, utilizing the Spearman to pinpoint core keywords, we assess the linear and nonlinear correlations between search keywords and international crude oil prices. Finally, PCA is employed to reduce the dimensionality of network search data. PCA is a commonly used dimensionality reduction technique that reduces the dimensionality of data while preserving as much original information as possible. When processing high-dimensional search index data, PCA can not only minimize data dimensions, mitigate multicollinearity effects, and reduce data pair complexity, but also filter out noise in the data and preserve as much original information as possible. Compared to other dimensionality reduction methods such as linear discriminant analysis and independent component analysis, PCA does not require category labels for data and is suitable for unsupervised learning [58]. Hence, PCA is simple, computationally efficient, and ideal for handling massive amounts of data.

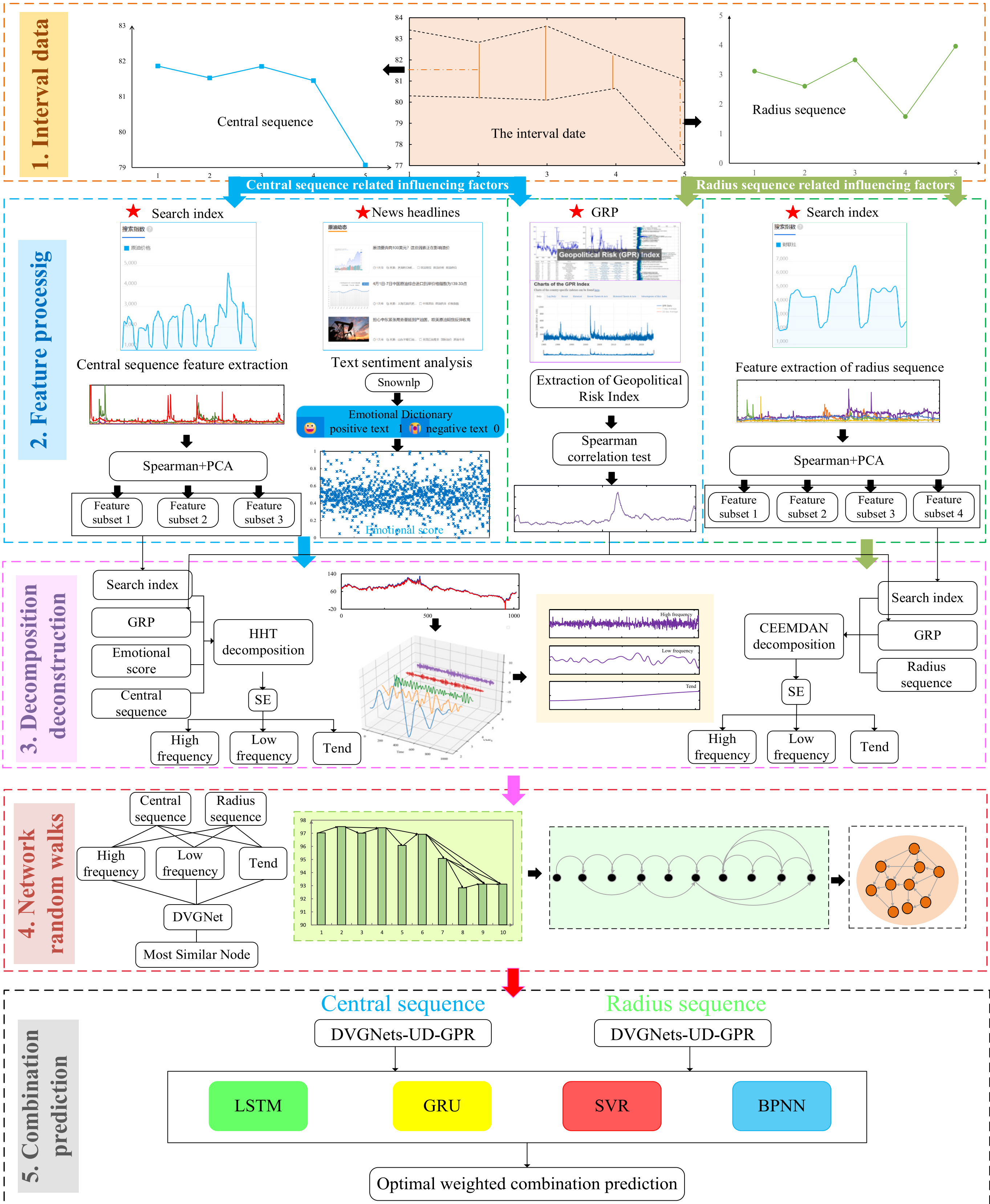

Fig. 3. Modeling flowchart.

2.3.2. News headline sentiment analysis

The crude oil market, influenced by a cascade of news or significant events, demonstrates a heightened sensitivity to qualitative factors, particularly regarding price fluctuations. In addition, news headlines on social media may be more credible and persuasive than other content, as those headlines contain less noisy information. Therefore, we systematically gathered daily news items pertaining to global crude oil prices from websites that focus on international crude oil news. Owing to the presence of substantial noise in the text, which includes symbols and punctuation, preprocessing is necessary. This process involves data cleansing, segmentation, stop word removal, and conversion into textual data for subsequent analysis. We will utilize the Snownlp toolkit in Python to conduct sentiment analysis on the preprocessed text. This task will involve calculating the sentiment value of each news text based on a sentiment corpus. The sentiment score ranges from 0 to 1, where a score closer to 1 indicates a more positive sentiment. The sentiment value obtained reflects the emotional tendency expressed by the news title text.

2.4. Data decomposition

Time series of international crude oil prices exhibit characteristics such as non-stationarity, significant randomness, pronounced chaos, and multi-scale variability due to the impact of uncertain factors. Typically, these time series comprise various frequency components, including trends, periodic patterns, seasonal fluctuations, and random fluctuations. Hence, employing decomposition methods becomes imperative to make time series more regular and stable. HHT decomposition and CEEMDAN are more robust compared to the commonly used EMD decomposition, especially in dealing with nonlinear and non-stationary signals. They can handle possible problems that may arise during EMD decomposition, such as mode aliasing and end effects. Therefore, this paper uses HHT decomposition to analyze the center and its eigenvalues and applies CEEMDAN decomposition to analyze the radius and its eigenvalues.

The HHT decomposition aims to analyze signals that are both nonlinear and non-stationary. Primarily, it comprises two key components: EMD and Hilbert transform. For the EMD decomposition obtained component $h_i(t)$, perform a Hilbert transform to obtain the complex form of the component $z_i(t)$: $z_i(t) = h_i(t) + jH[h_i(t)]$, which $H[\cdot]$ represents the Hilbert transform and j is an imaginary unit.

EEMD adds Gaussian white noise based on EMD to solve the modal aliasing problem, while CEEMDAN improves on EEMD by adding specific adaptive noise at each stage of decomposition. Compared with EEMD, it improves the calculation speed and obtains better modal decomposition results. The decomposition process of the CEEMDAN algorithm is as follows:

Step 1: Add Gaussian white noise $\omega_i(t)$ of ε_0 amplitude to the signal $x(t)$,$x(t)' = x(t) + \varepsilon_0\omega_i(t)$, where i is the number of times white noise is added.

Step 2: Defined $E_j[\cdot]$ as the jth intrinsic mode component after EMD decomposition, perform i times of EMD decomposition on $x(t)'$, and take the mean of the results to obtain the first intrinsic mode component $IMF_1(t)$,

$$E_j[x(t)'] = IMF_1^i(t) \tag{2}$$

$$IMF_1(t) = \frac{1}{I}\sum_{i=1}^{I} IMF_1^i(t) \tag{3}$$

Step 3: Calculate the residual $R_1(t)$ after removal $IMF_1(t)$,

$$R_1(t) = x(t)' - IMF_1(t) \tag{4}$$

Step 4: Calculate the residual $R_2(t)$ of the second intrinsic mode component $IMF_2(t)$,

$$IMF_2(t) = \frac{1}{I}\sum_{i=1}^{I} E_1[R_1(t) + \varepsilon_i E_1[\omega_i(t)]] \tag{5}$$

$$R_2(t) = R_1(t) - IMF_2(t) \tag{6}$$

Step 5: For other stages, repeat the above steps to calculate the k intrinsic mode component and its residual.

$$IMF_k(t) = \frac{1}{I}\sum_{i=1}^{I} E_1[R_{k-1}(t) + \varepsilon_{k-1}E_{k-1}[\omega_i(t)]] \tag{7}$$

$$R_k(t) = R_{k-1}(t) - IMF_k(t) \tag{8}$$

The original signal subsequence obtained by decomposition using the above method has a specific fluctuation pattern, and the fluctuation patterns between partial molecular sequences are similar. To consider short-term fluctuations and long-term trends of data and enhance the understanding of multi-scale data, it is necessary to reconstruct the data into high-frequency, low-frequency, and trend terms. The SE can be calculated by adjusting the lag window size to calculate the entropy value of the data sequence and then extracting information for reconstruction. Selecting sample entropy as a method for reconstructing subsequence data into high frequency, low frequency, and trend is mainly due to its ability to capture nonlinear features of the data, avoid strict data assumptions, adapt to multi-scale analysis, and fit in practical applications. Compared to traditional linear hypothesis methods, SE can better meet the needs of complex data analysis and provide deeper data understanding and application value.

2.5. Network random walks

Link prediction is an important problem in network science since it includes predicting when two disconnected nodes will form a link or inferring missing links in the network, that is, predicting unknown nodes by analyzing existing node information in the network. In networks, a popular link prediction strategy is to use random walk processes to assess the possibility of a node connecting to another node by giving probability scores to a group of nodes [59]. Converting time series into networks and using network analysis techniques for prediction can be seen as an application of link prediction, which uses link prediction techniques to discover periodic or nonlinear relationships in time series that may not have appeared previously but may appear in the future. In summary, by transforming time series data into a network structure for link prediction, link prediction techniques can more accurately identify the historical time points that exhibit the highest correlation with the current time, and use them as feature values to predict the next connected node. Therefore, based on this method, the prerequisite for prediction is to first convert the time series into a directed visibility graph network that can reflect the characteristics and traits of the time series. Calculating node similarity allows us to assess the similarity of network nodes and understand the impact of history and present data on the future.

The visibility graph algorithm proposed by Lacasa et al. [40] can map time series into the network under certain connection conditions, and the visible rays between data determine the connection of nodes. For any two nodes $A(x_i, y_i)$ and $B(x_j, y_j)$, a point $C(x_k, y_k)$ always lie between those two points. When $x_i < x_k < x_j$ the equation satisfies:

$$y_k < y_j + (y_i - y_j)\frac{x_j - x_k}{x_j - x_i} \tag{9}$$

Nodes A and B are visible to each other in the network and can be directly connected as shown in Fig. 4. Undirected networks are transformed into directed networks as a result of the irreversibility of time series; nodes with smaller values point to nodes with larger values. The graph contains 10 different values, with the horizontal axis representing

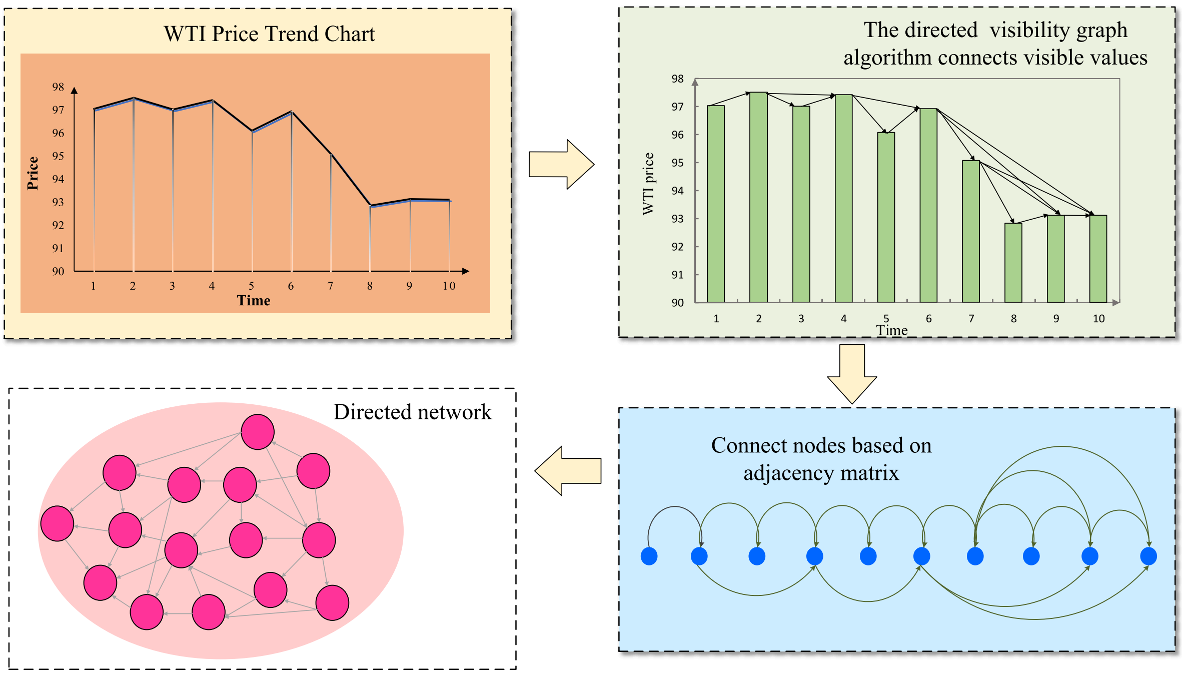

Fig. 4. Map time series into Directed Visibility Graph Networks.

the time point. Each value is represented by a bar graph at its corresponding position. Then, the vertex of each bar graph is connected to the vertex positions of other bar graphs that can be seen at its vertex position in chronological order using directed rays. Finally, the time point is regarded as a node, and the directed rays are retained to obtain the directed network. Each time series can obtain a unique visibility graph network, regardless of the way how the bar graph is constructed. The data is calculated using an improved directed visible graph algorithm to obtain the adjacency matrix, and the directed visibility network is drawn through the adjacency matrix.

The similarity of link nodes is computed using the local random walk algorithm after acquiring the directed network, and the transfer matrix P is used to represent the random walk process. In a network with N nodes, the probability of a walker transversing from node i to node j is denoted by P_{ij}:

The transition matrix from node i to node j is:

$$Q_{ij} = \begin{cases} \frac{a_{ij}}{k_i^{out}}, & if(i,j) \in E \\ 0, & otherwise \end{cases} \tag{10}$$

Among them, $A = (a_{ij})_{N\times N}$ is the adjacency matrix of the directed network. The above formula a_{ij} represents the values corresponding to the i^{th} row and j^{th} column of the adjacency matrix corresponding to the directed network. In a directed network, if node i and node j are connected, then $a_{ij} = 1$, otherwise $a_{ij} = 0$, k_i^{out} represents the out-degree of node i.

The process will be iterated till the walker moves through the whole network after t steps. Then the probability, written as $\vec{\pi_i}$:

$$\vec{\pi_i}(t) = Q^T \vec{\pi_i}(t-1) \tag{11}$$

In the above equation, Q^T is the transposition of the matrix Q. In the initial state, when $t = 0$, $N \times 1$ the row vector, denoted by $\vec{\pi_i}(0)$, is used to describe the random walker standing at the node i. The i^{th} element $\vec{\pi_i}(0)$ equals 1 and the others equal 0, representing that the walker starts to walk from the node i.

Calculate the similarity between node i and node j:

$$S_{ij}{}^{LRW}(t) = \frac{k_i}{2|E|}\pi_{ij}(t) + \frac{k_i}{2|E|}\pi_{ji}(t) \tag{12}$$

The above equation $|E|$ suggests the total number of node edges in the network, k_i represents the degree of node i, and k_j refers to the degree of node j. In the procedure of going from node i to node j, the walking may take the walker too far away from the goal node. In this scenario, downstream walking tends to confine itself to the local vicinity rather than exploring the entire network, potentially leading to reduced prediction accuracy. To enhance the degree of similarity between the target node and its adjacent nodes, it is crucial to aggregate the results of individual random walks. The final result of the stacking operation is an increase in the similarity between node i and node j.

$$S_{ij}{}^{SRW}(t) = \sum_{l=1}^{t} S_{ij}{}^{LRW}(l) \tag{13}$$

SRW in the preceding equation stands for a stacked random walk. Consequently, the similarity between node i and node j is obtained, and the similarity between all nodes in the network is obtained. In essence, to predict the value of X_{n+1} nodes, it is imperative to initially identify nodes with high similarity to X_n nodes for prediction. These nodes with high similarity will serve as predicted feature values.

2.6. Interval combination forecasting

A single model may not be the most suitable choice for fitting multi-frequency patterns in interval subsequences, as it can only capture specific data features [54]. For this reason, we chose four machine learning and deep learning methods for prediction (BPNN, SVR, LSTM & GRU). First, as a classic neural network model, BPNN possesses a high degree of adaptability and non-linear learning ability. It is capable of processing a large amount of data in parallel to capture the random characteristics of multivariate time series, and is suitable for predicting time series with large fluctuations. Secondly, SVR possesses a robust capacity to manage nonlinear relationships in data. SVR can capture complex patterns in the data by mapping complex nonlinear data to high-dimensional space for linear processing, which is achieved through the use of kernel functions. Thirdly, the LSTM is a deep learning model that can selectively filter information through a "gate" structure to effectively mine deep features of the time series of crude oil prices. For low-frequency time series with cyclic patterns resembling repeated cycles, LSTM can better capture these periodic features. Fourthly, GRU lowers memory and computational costs while maintaining a strong capture of long-term dependencies. Therefore, we combined the benefits of four methods to predict the high-frequency, low-frequency, and trend of the center and radius, achieving multi-angle capture of modal features. The specific steps are shown in Fig. 5:

Step 1: Input the relevant characteristic values of the high-frequency, low-frequency, and trend items of the center into the four prediction methods mentioned above for separate predictions, and then calculate the predicted values of the high-frequency, low-frequency, and trend items of the center under the four different prediction methods.

Step 2: Input the relevant characteristic values of the high-frequency, low-frequency, and trend terms of the radius into the four prediction methods mentioned above for separate predictions, and obtain the predicted values of the high-frequency, low-frequency, and trend of the radius under the four different prediction methods.

Step 3: Accumulate the high-frequency, low-frequency, and trend predicted values of the center predicted by GRU to obtain the center predicted value of GRU. Then, add up the high-frequency, low-frequency, and trend predicted values of the radius predicted by GRU to obtain the radius predicted value of GRU, and so on to obtain the center predicted value and radius predicted value under the other three methods.

Step 4: Calculate the optimal center and radius values for the four methods using the optimal weighting method.

3. Empirical analysis

This section comprises a series of experiments that were performed to authenticate the efficacy of the suggested model in forecasting global crude oil prices. First, introduce data sources to collect and organize unstructured data. Following this, the HHT and CEEMDAN methods will be used to decompose the center, radius, and multi-source information time series, respectively to regularize the subsequences. Then, map the center and radius subsequences into a network and select nodes with high similarity as feature values. Finally, combine and predict all of the interval subsequences to obtain the final prediction result.

3.1. Data preprocessing

3.1.1. Data collection

Crude oil boasts the world's most actively traded commodity futures with a long history. The prices of crude oil futures are a major economic and financial market indicator and a key player in the international energy industry. When it comes to energy futures markets, the New York Mercantile Exchange (NYMEX) has long been at the vanguard, particularly in the trading volume of WTI crude oil futures. Therefore, choosing the WTI crude oil futures price of NYMEX as the research object is fairly representative. The data was sourced from the website of the NYMEX (http://www.cmegroup.com/). The period spans from January 1, 2020, to November 21, 2023, and the price trend is shown in Fig. 6. The data before September 26, 2023, will be used as the training set, while the remaining portion will act as the predicted test set.

3.1.2. Internet search index

As the supplementary market data source, Baidu search can reflect how the general population feels about the global crude oil market. As a supplementary source of market data, the Baidu search index can to some extent reflect the general public's perception of the global crude oil market. Although Baidu's search mainly reflects the attention of investors in the Chinese market, it can provide some insight into global market sentiment by inputting combined appropriate keywords. First, this article selected phrases from international crude oil news headlines that have the potential to gain widespread attention globally, reflecting the level of global public and investor attention and emotional fluctuations towards these events. Secondly, as a major global consumer of crude oil, China's market sentiment also has a significant impact on international crude oil prices [60]. When major events or fluctuations occur in the international crude oil market, the search behavior of the Chinese public may surge dramatically. This change in search behavior can indirectly reflect the global market's response to these events. To be more specific, major geopolitical events, OPEC conference outcomes, or

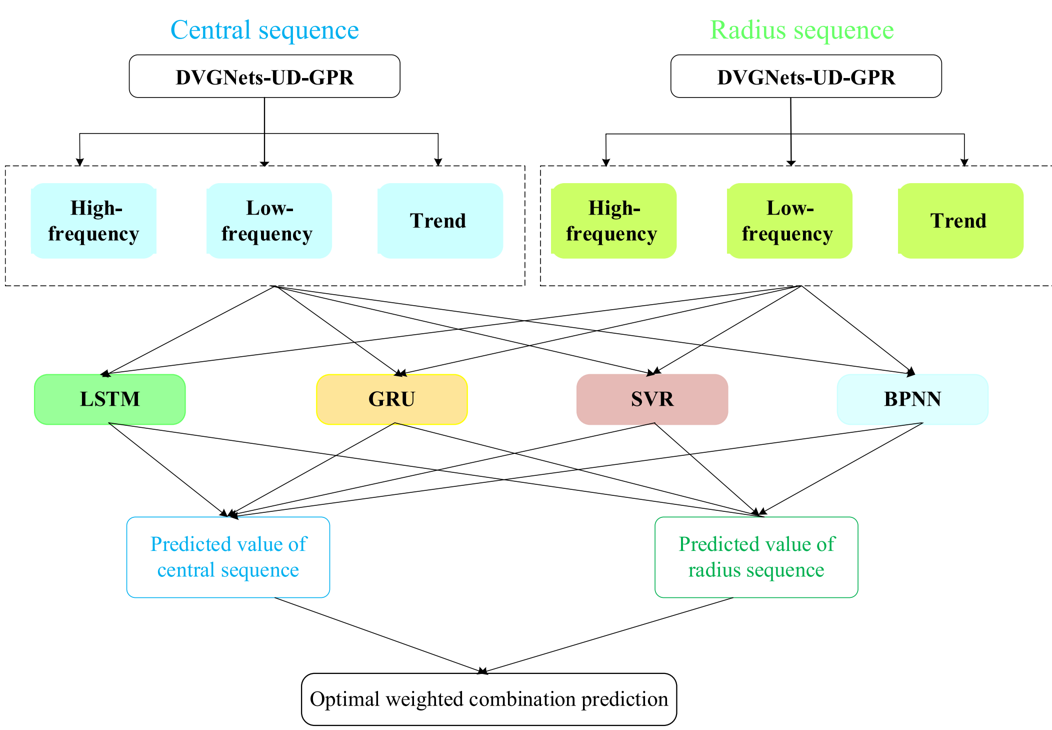

Fig. 5. Interval combination forecasting.

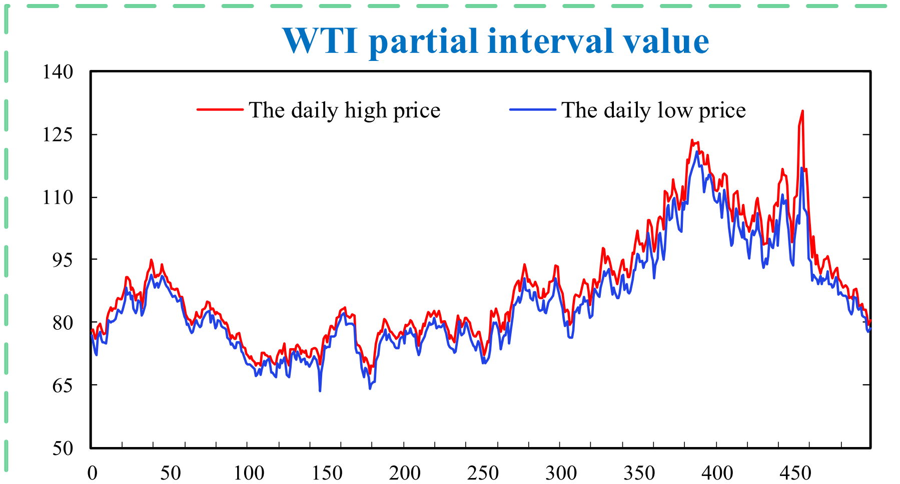

Fig. 6. WTI interval value data.

significant fluctuations in oil prices may all trigger an increase in search volume. The table below shows that most keywords such as "international crude oil prices", "OPEC", and "Fourth Middle East War" have been verified to correlate with international crude oil prices, which can reflect the market sentiment of global crude oil. As a result, the Baidu Index functions as a metric to indicate investors' search behavior regarding crude oil futures. The procedure for quantifying investor attention hotspots using the Baidu Index involves the following steps:

(1) Identify initial keywords based on the factors influencing international crude oil prices. Leverage headlines sourced from reputable oil websites that cover international crude oil news as a corpus to extract keywords reflecting the online search behavior of investors. First, crawl the relevant headlines, preprocess the data, divide the number of topics with the LDA model, and determine the top five topic keywords for each topic as candidate keywords. Finally, a total of 65 candidate keywords such as Caixin News Agency, US dollar index, and oil crisis are determined.

(2) The Spearman method was applied to investigate the correlations among the search keywords and the WTI center sequence and radius sequence. Moreover, thirty keywords exhibiting a strong correlation with the center sequence and another thirty keywords exhibiting a strong correlation with the radius sequence were chosen separately. The specific situation is shown in Table 1.

(3) PCA is used to reduce the dimensionality of core keywords and quantify the characteristics of trending topics that pique the interest of investors. Reduce the keyword data related to the central sequence to three dimensions, while the keyword data related to the radius sequence should be reduced to four dimensions. More precisely, select a new sequence with a cumulative contribution rate of 98% as the input for the prediction model.

Table 1
Selected keywords.

Radius keywords	Correlation	Central keywords	Correlation
Brent crude oil**	0.6707	Cailian Press**	0.6916
International crude oil prices**	0.6439	Oil prices**	0.6747
Oil prices**	0.5609	New energy**	0.6036
Crude oil futures**	0.5008	International crude oil prices**	0.5304
Cailian Press**	0.4761	New energy**	0.5277
Oil prices**	0.4749	Energy crisis**	0.4473
OPEC**	0.4513	PTA Futures**	0.4108
Energy crisis**	0.4472	Brent crude oil**	0.3952
Fuel**	0.4404	Fuel**	0.3272
Geopolitics**	0.4276	The Fourth Middle East War**	0.3217
New energy**	0.3877	Natural gas**	0.3061
Crude oil plummets**	0.3709	Coking coal**	0.2950
The oil crisis**	0.3403	Geopolitics**	0.2804
PTA Futures**	0.3330	Azerbaijan**	0.2534
Saudi Arabia**	0.3310	Oil tank truck**	0.2218
Refinery**	0.3149	Coal**	0.2000
The Fourth Middle East War**	0.3075	Drilling**	−0.2088
Finished oil price**	0.3057	Oilfield**	−0.2166
New energy**	0.2939	Shale**	−0.2177
Iraq**	0.2911	Gulf of Mexico**	−0.2642
Price adjustment mechanism for refined oil products**	0.2796	Price war**	−0.2781
Asphalt futures**	0.2663	Mexican oil**	−0.3018
Energy**	0.2507	RMB to USD exchange rate**	−0.3173
Jamaica system**	0.2443	Domestic refined oil price adjustment window**	−0.3393
Global economic situation**	0.2229	Drilling rig**	−0.3486
US dollar index**	0.2208	LED color screen**	−0.3505
No renewable energy**	0.2165	Beck**	−0.3992
Coal**	0.2113	Market trends in the Shanghai and Shenzhen stock markets**	−0.4274
UAE**	0.2076	Oil riots**	−0.4444
Strategy**	0.2073	The United States**	−0.5011

Note: *, **, and *** respectively represent significance levels of 10%, 5%, and 1%.

3.1.3. Sentiment analysis

The news text of the crude oil market reflects the potential emotional tendencies and decision-making behaviors of investors. By identifying and summarizing emotional vocabulary in the news text, it is possible to quantify investors' attitudes towards the text. This study crawled news headlines and retrieved text data from worldwide petroleum websites that covered international crude oil news headlines between January 1, 2020, to November 23, 2023. The original news text data obtained through crawler tools has a complex structure, containing a large amount of invalid information and noise factors. Therefore, this article needs to perform data cleaning and word segmentation processing.

(1) First, clean the data. Delete duplicate text that appears during the news crawler process due to network and other factors. Simultaneously, the news article's headline is a condensed and polished version of its substance. Thus, this article solely considers

the material under the "News Title" field while excluding all other dimensions. Following that, based on the Chinese stop word list file, the news title is text cleaned to remove unneeded punctuation marks like quotation marks, colons, and parentheses, as well as noise strings that interfere with semantics, making the title text concise and effective in reflecting emotional tendencies.

(2) Secondly, calculate the sentiment value after word segmentation. Use jieba segmentation code to process the word segmentation of news headlines that have undergone data cleaning, to reduce the coupling of context and the impact of word order. After data cleansing and word segmentation processing, use the Snownlp database for sentiment analysis on pre-processed news title texts to obtain corresponding sentiment values. To get the daily average sentiment score, the sentiment scores of numerous comments received daily are averaged. Missing sentiment data is imputed with the mean method.

3.2. Data decomposition

The traditional prediction methods usually fail to capture the fluctuation frequencies of crude oil price time series given their non-linear and non-stationary nature. Hence, decomposing the time series into multiple more regular and stable components becomes necessary to mitigate data complexity. This paper uses the HHT decomposition method to decompose the WTI crude oil center sequence and its related influencing factors. The CEEMDAN method is utilized to decompose the

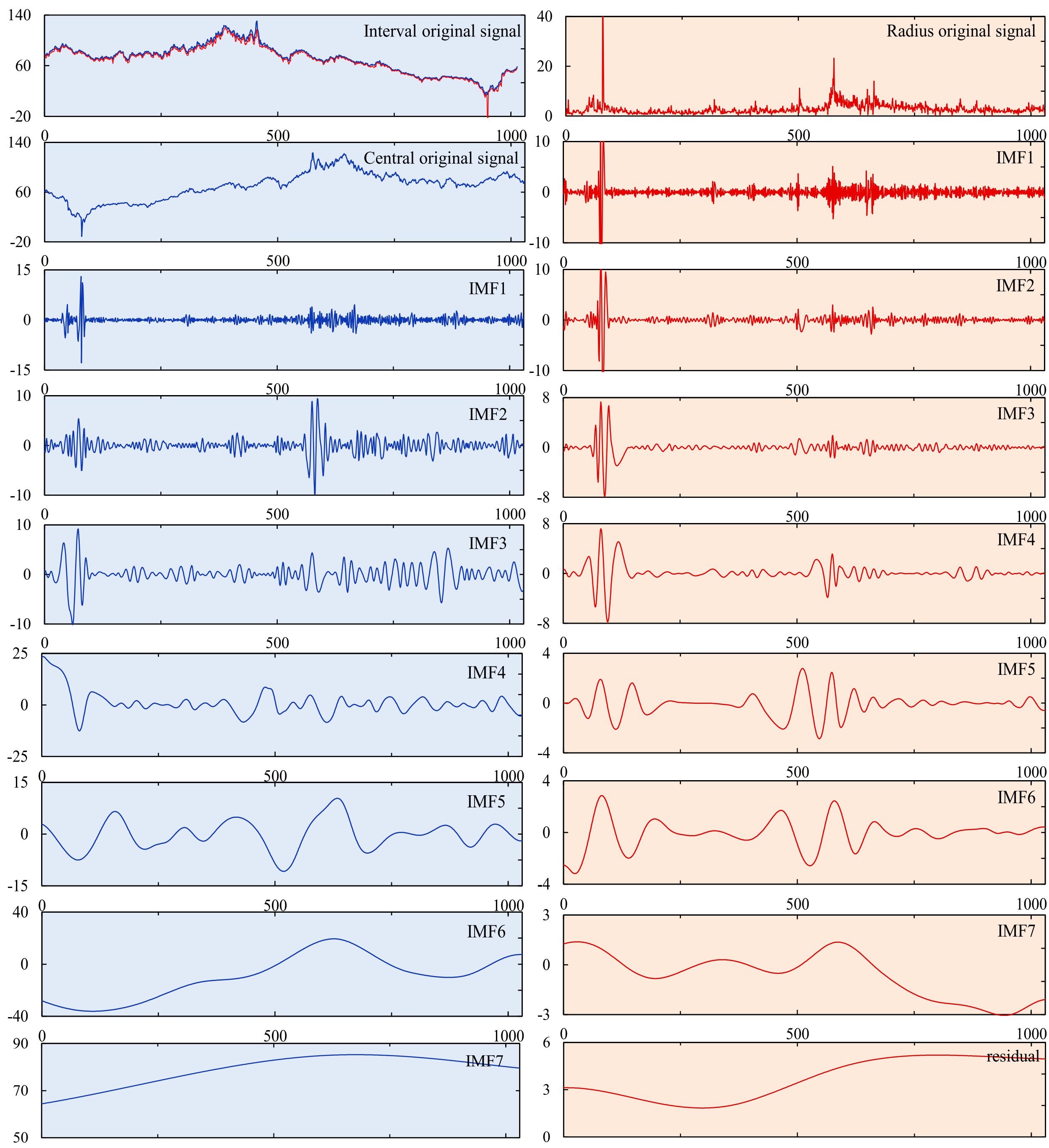

Fig. 7. Decompose WTI interval data.

WTI crude oil radius sequence and its related influencing factors. The decomposed center sequence and radius sequence are shown in Fig. 7. The prediction model will become excessively complex and cumbersome if each sequence is explicitly predicted, as data decomposition results frequently contain multiple sequences. Therefore, this paper reconstructs the results of HHT decomposition and CEEMDAN decomposition. First, input the decomposed sequence into the SE model to determine the entropy value of each sequence. Secondly, choose sequences with similar and relatively high entropy values to superimpose into the high frequency of the original sequence, sequences with similar and relatively low entropy values to superimpose into the low frequency, and the remaining sequences to superimpose into the trend term. The reconstructed sequence is shown in Fig. 8–9.

3.3. Mapping time series into the network and searching similar nodes

For the high-frequency, low-frequency, and trend components of the decomposed center sequence and radius sequence, the DVGNets are used to map these six subsequences into directed networks. First, use the visible graph algorithm to calculate the adjacency matrix of the subsequence x_i. Subsequently, the data is woven as nodes into a directed network $G_i(V,E)$ through the adjacency matrix. In the constructed network, the data values are encoded as nodes V, and the connection between any two points is E.

Once the network associated with each modal component is obtained, a local random walk algorithm is used to calculate the similarity between nodes in the network, generate a similarity matrix between nodes, and identify the similarity values between other nodes and the last node. The first few nodes with the highest similarity are selected as the final feature values and input into the prediction model.

3.4. Combination forecasting results

Following the organization of relevant data, the relevant influencing factors of the center and radius sequences, as well as the filtered similar node values, are employed as input data. GRU, LSTM, SVR, and BPNN are selected for prediction, and the corresponding center and radius subsequence prediction values are obtained using the four methods. Subsequently, utilizing the optimal weighted combination method, the four predicted values are aggregated to yield the optimal WTI crude oil price forecast. Based on the interval prediction results depicted in Fig. 10, we can observe a strong alignment with the actual price fluctuation trends. This indicates a commendable fit between the model and the real-world data, suggesting a high level of predicted accuracy. Thus, the model demonstrates robust performance in forecasting, providing valuable insights into and effective mitigation strategies for market volatility. (Note: The data in Fig. 11 ranges from 1 to 40, representing the period from November 21, 2023 to September 27, 2023.)

4. Method evaluation

After the proposal of evaluation indicators for the experimental results in this section, four sets of controlled trials are utilized to illustrate the superiority of our method.

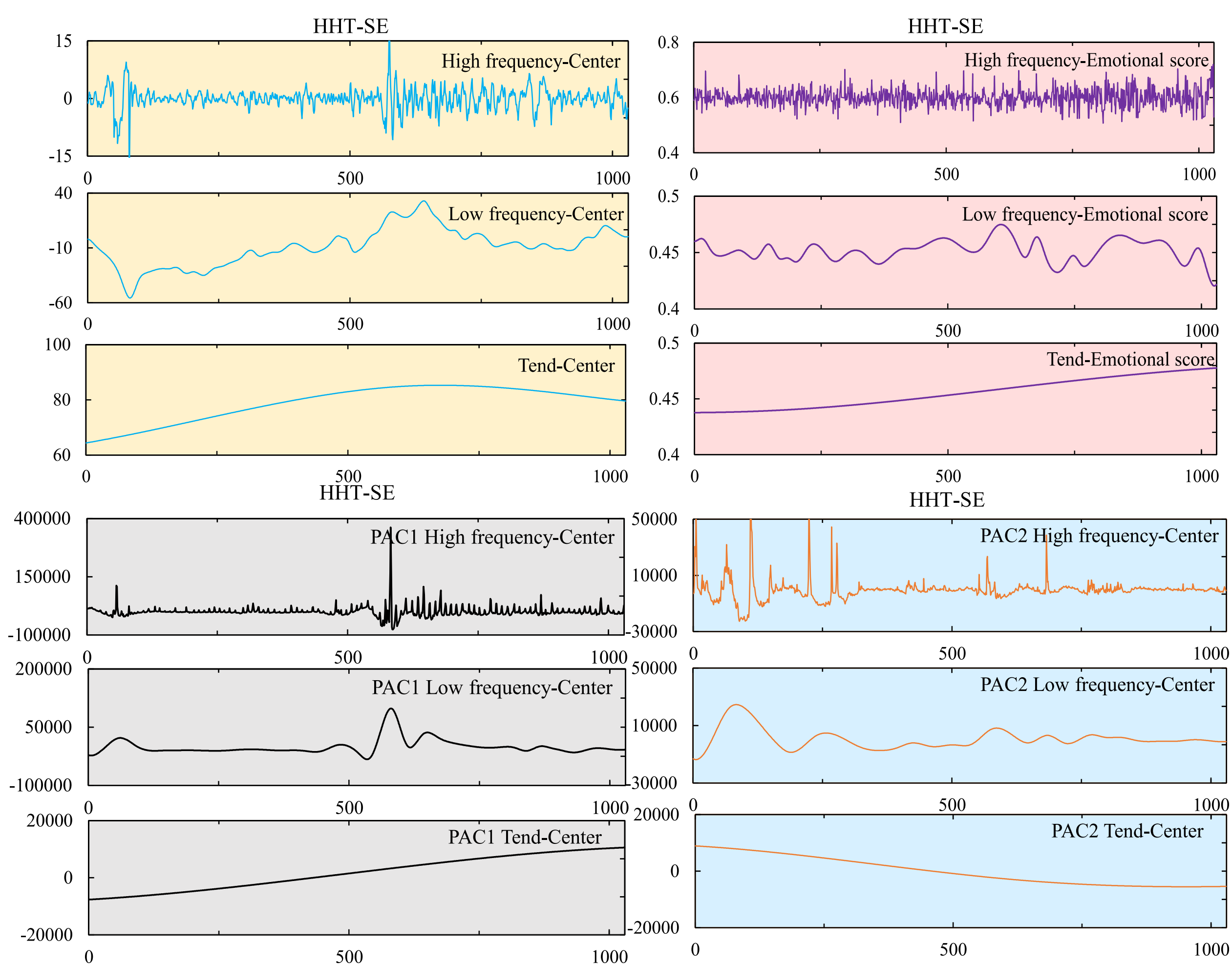

Fig. 8. Partial HHT decomposition and reconstruction of data.

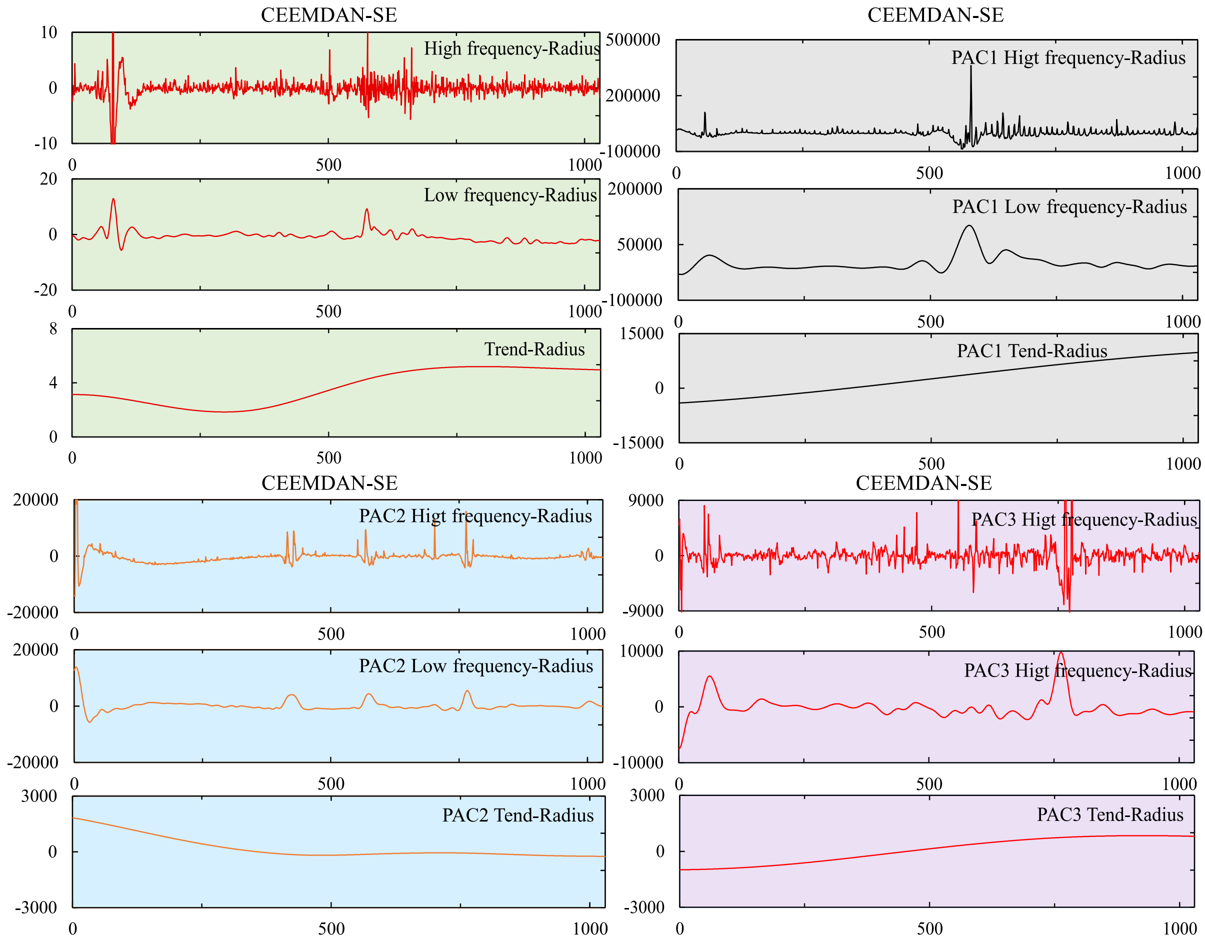

Fig. 9. Partial CEEMDAN decomposition and reconstruction of data.

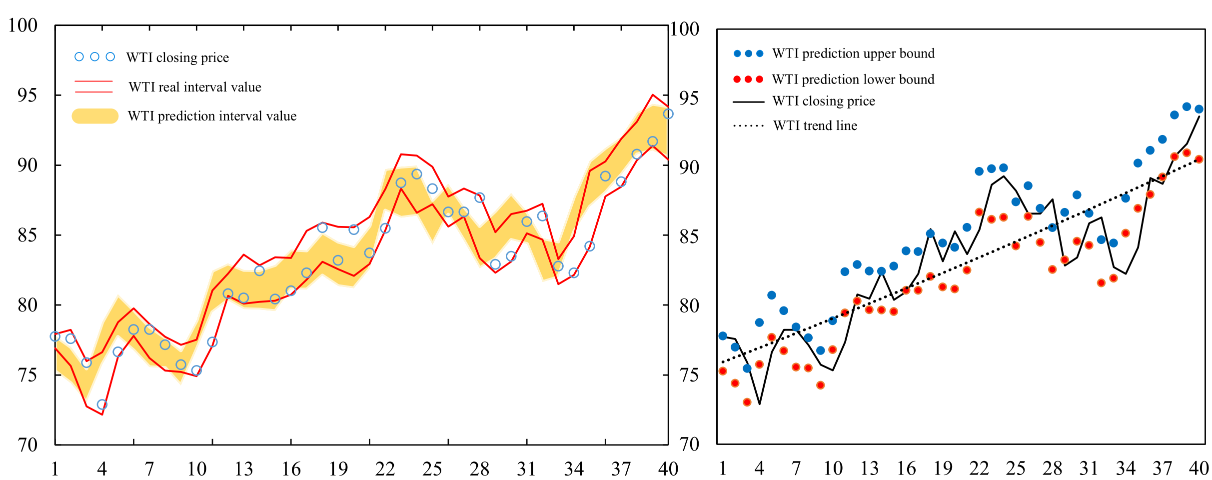

Fig. 10. Forecasting results of the proposed model.

4.1. Experimental design

To assess the efficacy of the proposed method in this study, four error evaluation metrics were employed: Interval Mean Absolute Percentage Error (IMPE), Interval Mean Relative Error (IARV), Root Mean Square Error (IRMSE), and Interval Mean Distance Error (IMDE). These metrics were utilized to assess the effectiveness of the prediction method outlined in this paper.

$$IMAPE = \frac{1}{2n}\sum_{i=1}^{n}\left(\left|\frac{x_i-\widehat{x}_i}{x_i}\right|+\left|\frac{r_i-\widehat{r}_i}{r_i}\right|\right) \tag{14}$$

$$IARV = \frac{1}{n}\left(\frac{\frac{1}{n}\sum_{i=1}^{n}\left(Var(\widehat{x}_i)-Var(x_i)\right)^2}{Var(x_i)}+\frac{\frac{1}{n}\sum_{i=1}^{n}\left(Var(\widehat{r}_i)-Var(r_i)\right)^2}{Var(r_i)}\right) \tag{15}$$

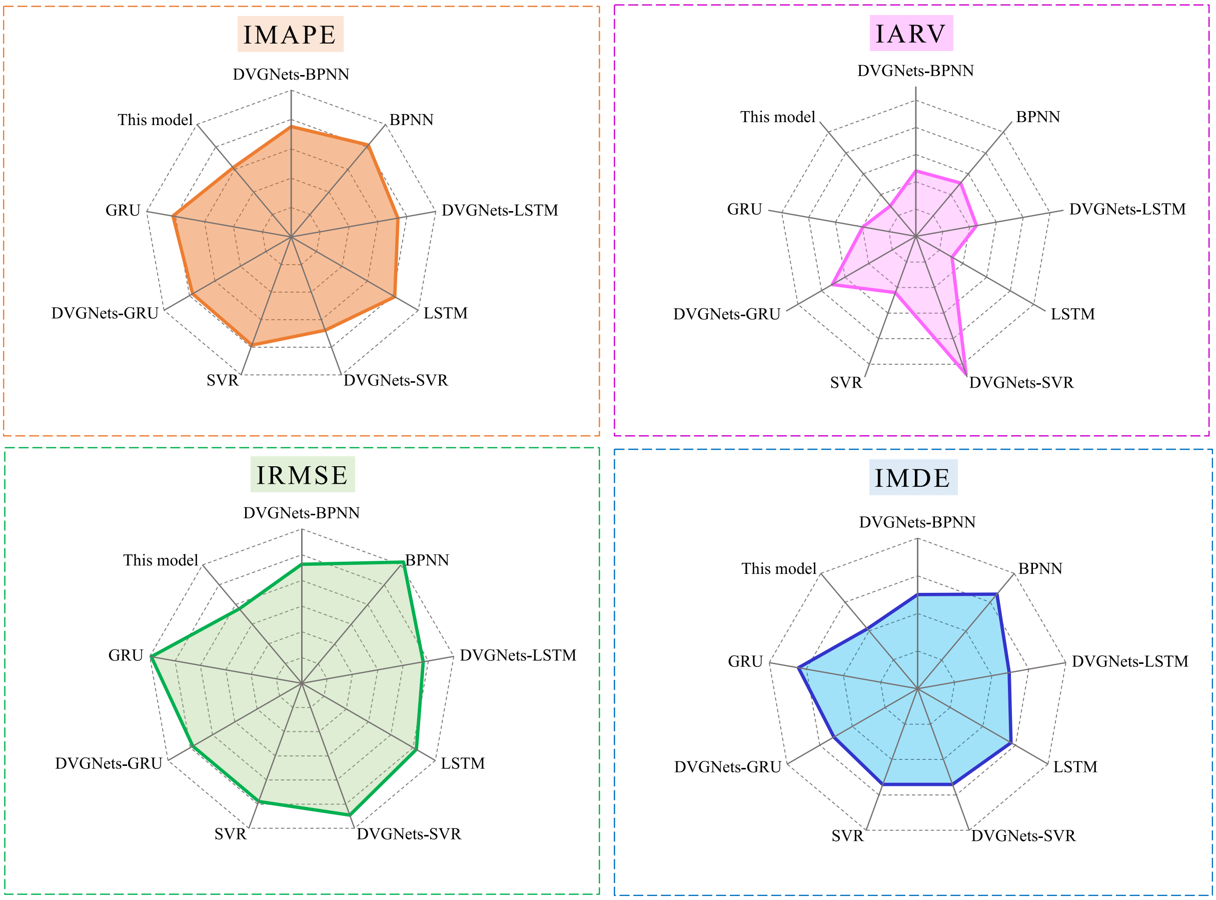

Fig. 11. Comparison of predictive power in experiment 1.

$$IRMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n}\left((x_i - \widehat{x}_i)^2 + (r_i - \widehat{r}_i)^2\right)} \tag{16}$$

$$IMDE = \frac{1}{2n}\sum_{i=1}^{n}(|x_i - \widehat{x}_i| + |r_i - \widehat{r}_i|) \tag{17}$$

Among them, n refers to the number of observed values, x_i is the actual center value, $\widehat{x}_i$ indicates the predicted center value, r_i means the actual radius value, $\widehat{r}_i$ stands for the predicted center value, and $Var[\cdot]$ denotes the variance.

To further determine whether the improvement in prediction accuracy of the proposed model compared to other models is statistically significant. The Diebold Mariano (DM) test, Wilcoxon signed rank test, and The Paired t-test methods will be used to test the superiority of the proposed model.

(1) The Diebold Mariano (DM) test

Assuming there is no significant difference in predictive performance between prediction models. If the DM statistic is less than 0, it indicates that the average prediction error of Model A is smaller than that of Model B, which means that Model A is better than Model B [61]. If the calculated is less than 0.05, it indicates that Model A is significantly better than Model B. The definition of DM statistics is:

$$\mathrm{DM} = \frac{\overline{d}}{\sqrt{\frac{1}{n-1}\sum_{i=1}^{n}(d_i - \overline{d})^2}\Big/\sqrt{n}} \tag{18}$$

Among them, $d_i = (|x_i - \widehat{x}_{Ai}| + |r_i - \widehat{r}_{Ai}|) - (|x_i - \widehat{x}_{Bi}| + |r_i - \widehat{r}_{Bi}|)$, $\overline{d} = \frac{1}{n}\sum_{i=1}^{n} d_i$, $\widehat{x}_{Ai}$ and $\widehat{x}_{Bi}$ are the center predicted values of model A and model B at time i, $\widehat{r}_{Ai}$ and $\widehat{r}_{Bi}$ are the radius predicted values of model A and model B at time i. This study uses a one-sided t-test to determine whether model A has an advantage over model B.

(2) The Wilcoxon signed-rank test

Wilcoxon signed-rank test is a no-parametric statistical test method that can be used to compare the superiority of two prediction models in terms of prediction accuracy. By applying statistical tests to the differences$d_{iw} = (x_i - \widehat{x}_{Ai})^2 + (r_i - \widehat{r}_{Ai})^2 - \left((x_i - \widehat{x}_{Bi})^2 + (r_i - \widehat{r}_{Bi})^2\right)$ between the predicted results of the model, determine whether the differences are significant. When the mean prediction error $\overline{d}_w$ between Model A and Model B is less than 0, it indicates that Model A is superior to Model B. If the p-value of the test result is less than 0.05, it indicates that Model A performs significantly better than Model B in a statistical sense, indicating that the model has better predictive ability in the evaluated indicators.

(3) The Paired t-test

Paired t-test is a statistical method used to compare the mean differences between two groups of related samples. For interval prediction models, the performance of the model can be evaluated by comparing the prediction errors of two models on the same set of observations. For models A and B, calculate the difference in prediction error $d_{ip} = \sqrt{(x_i - \widehat{x}_{Ai})^2 + (r_i - \widehat{r}_{Ai})^2} - \sqrt{(x_i - \widehat{x}_{Bi})^2 + (r_i - \widehat{r}_{Bi})^2}$ for each sample point separately. Then perform paired t-tests to compare whether there is a significant difference between the two models. When the mean prediction error $\overline{d}_p$ is less than 0, Model A's prediction performance is better than Model B. When the significance level p-value is less than 0.05, it indicates that Model A is significantly better than Model B.

4.2. Contrast experiment

This section demonstrates the superiority of the proposed combination model through three sets of experiments and comparative analysis. The first experiment (as can be seen in Section 4.2.1) examined the predictive performance of models with and without DVGNets included. The second experiment (as can be seen in Section 4.2.2) seeks to illustrate the real-time influence of multi-source information on prediction. Meanwhile, the third experiment (as can be seen in Section 4.2.3) evaluates the efficacy of the optimal weighted combination method in enhancing the fitting performance of various models. The error evaluation results from the comparative experiment are presented in multiple tables, where lower values signify the improved predictive capability of the method.

4.2.1. Comparative link prediction of complex network

This study proposes a new method for predicting crude oil prices that utilizes DVGNets to identify node similarity and make related predictions, which is significantly different from traditional methods that use lag time for prediction. Table 2 and Fig. 12 display the comparative results of BPNN for predicting crude oil interval prices with and without DVGNets. The experiment suggests that the addition of DVGNets substantially reduces prediction errors, resulting in a higher level of consistency between the projected outcomes and the actual observed results. To mitigate the impact of random factors, LSTM, SVR, and GRU models were employed to compare the performance of traditional prediction and linked prediction. The results indicate that the implementation of the DVGNets approach led to a reduction in prediction errors for the LSTM, SVR, and GRU models. Notably, the IMAPE and MDE exhibited a substantial decrease. Overall, employing the DVGNets method for link prediction yields superior results compared to conventional prediction methods.

4.2.2. Comparison with multi-source information

The second comparison aims to determine whether multi-source information about WTI improves its price prediction performance. UD represents the impact of adding an Internet search index and sentiment analysis. According to the results in Table 3, compared to DVGNets-BPNN, DVGNets-UD-GPR-BPNN incorporates UD and GPR can bring out smaller prediction errors. Furthermore, after comparing three different machine learning approaches, it can be seen from a global perspective that the incorporation of UD improves prediction accuracy. For example, from the results in the chart, it is evident that after adding UD, the performance of the DVGNets-UD-LSTM model is better than that of the DVGNets-LSTM model. The prediction results of the DVGNets-UD-SVR model are superior to those of the DVGNets-SVR model. After further adding GPR, the prediction accuracy of DVGNets-UD-GPR-BPNN and DVGNets-UD-GPR-SVR was higher. In addition, the DVGNets-UD-GPR-GRU model also excelled over the models without UD and GPR.

Table 2
Evaluation results for experiment 1.

Method	IMAPE	IARV	IRMSE	IMDE
DVGNets-BPNN	0.1504	0.9616	1.8504	1.0010
BPNN	0.1633	1.0217	2.4622	1.3115
DVGNets-LSTM	0.1476	0.9047	1.9168	0.9880
LSTM	0.1629	0.6133	2.0576	1.1468
DVGNets-SVR	0.1353	2.1555	2.1829	1.0825
SVR	0.1567	0.8797	1.9566	1.0828
DVGNets-GRU	0.1550	1.4154	1.9546	1.0257
GRU	0.1635	0.7903	2.3739	1.2845
This model	**0.1233**	**0.5829**	**1.5114**	**0.8317**

Therefore, the Internet search index, news headline sentiment, and geopolitical risk indicators considered in this study as prediction eigenvalues can effectively improve the accuracy of crude oil price prediction.

4.2.3. Comparison with an optimally weighted combination

The third experiment is to assess the effectiveness of the optimal weighted combination method. The model in this section takes into account the influence of combination prediction methods. First, given the variety of machine learning algorithms, it is imperative to determine a suitable prediction method. According to the prediction results in Table 4 and Fig. 13, we have comprehensively explored eight common machine learning prediction algorithms, namely BPNN, SVR, LSTM, GRU, LASSO, CNN, Bayesian Regression (BR), and extreme learning machine (ELM). The results reveal that the application of BPNN, SVR, and LSTM greatly reduces prediction errors while improving fitting accuracy. Furthermore, compared to other evaluated algorithms, GRU can achieve the lowest IARV with outstanding performance. Therefore, we chose BPNN, SVR, LSTM, and GUR for the final optimal weighted combination prediction. For the prediction of the center value, we obtained weights of 0.004, −0.1493, 1.3946, and − 0.2554 for the four predicted values, respectively. For the prediction of radius values, we obtained weights of 0.4723, 0.0093, 0.1615, and 0.352 for four predicted values. The final prediction result was obtained by calculating the weights, and the four evaluation indicators showed that weighted prediction greatly improved accuracy. In comparison, it was found that this model provided the best fitting result.

In summary, the comparative analysis of the preceding three portions shows that our method can deliver excellent results in terms of prediction accuracy and precision.

4.3. Results of significance testing method

According to the compared model in 4.2 and the error evaluation of the proposed model, it can be seen that this model exhibits the best predictive performance in IMAPE, IRMSE, and IMDE. The Diebold Mariano (DM) test, Wilcoxon signed rank test, and Paired t-test results in Table 5 further indicate that the predictive performance of this model is much superior to other compared models in most cases, confirming its applicability and stability.

In the experiment, we refer to the model proposed in this article as Model A and other comparative models as Model B. According to the comparison results in Table 5, it can be seen that under the three testing methods, the values of DM, $\overline{d}_w$ $\overline{d}_p$ are all less than 0, indicating that the predictive performance of the proposed model in this article is superior to the compared model. First, the proposed model has significant advantages over traditional SVR, LSTM, GRU, and BPNN models that use hysteresis to predict crude oil interval prices, demonstrating that transforming crude oil price interval sequences into complex networks for link prediction is an effective prediction method. In addition, the proposed prediction method is significantly superior to four models: DVGNets-LSTM, DVGNets-BPNN, DVGNets-SVR, and DVGNets-GRU. Although link prediction was used, the prediction performance was still inferior s than the proposed method due to the lack of consideration of multiple sources of information. This reflects that adding these multiple sources of information to the prediction model can improve the sensitivity of the model to changes in crude oil prices. Finally, there will be major disparities in the prediction performance among different machine learning and deep learning methods. As indicated in the table, the SVR, BPNN, LSTM and GRU selected in this paper exhibit fewer prediction errors compared to other artificial intelligence methods. Therefore, by integrating the predicted values of these models through the optimal weighted average, the optimal prediction performance was obtained.

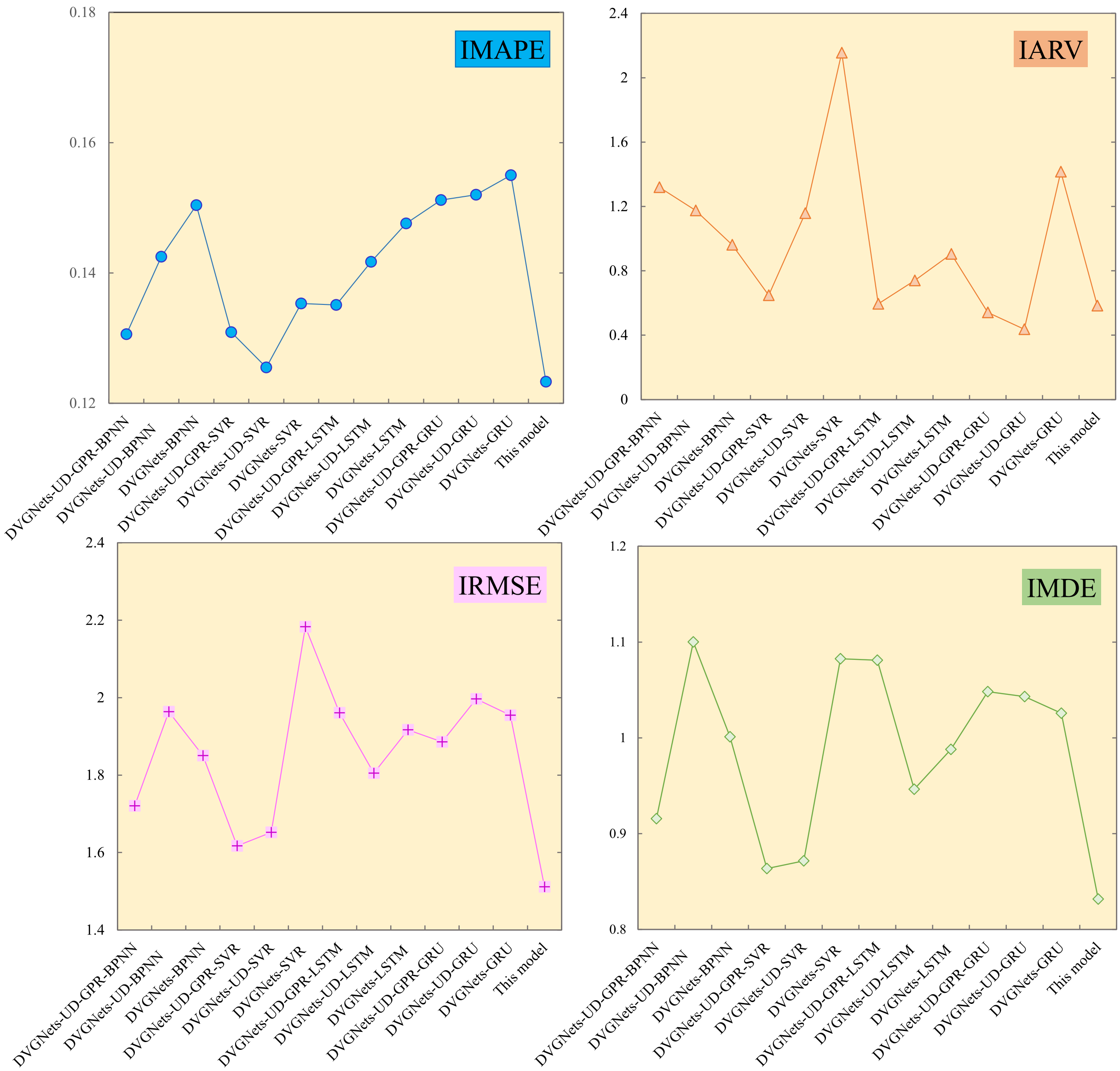

Fig. 12. Comparison of predictive power in experiment 2.

Table 3
Evaluation results for experiment 2.

Method	IMAPE	IARV	IRMSE	IMDE
DVGNets-UD-GPR-BPNN	0.1306	1.3190	1.7207	0.9156
DVGNets-UD-BPNN	0.1425	1.1755	1.9635	1.1002
DVGNets-BPNN	0.1504	0.9616	1.8504	1.0010
DVGNets-UD-GPR-SVR	0.1309	0.6483	1.6172	0.8636
DVGNets-UD-SVR	0.1255	1.1583	1.6523	0.8715
DVGNets-SVR	0.1353	2.1555	2.1829	1.0825
DVGNets-UD-GPR-LSTM	0.1351	0.5942	1.9608	1.0810
DVGNets-UD-LSTM	0.1417	0.7401	1.8051	0.9464
DVGNets-LSTM	0.1476	0.9047	1.9168	0.9880
DVGNets-UD-GPR-GRU	0.1512	0.5410	1.8859	1.0483
DVGNets-UD-GRU	0.1520	0.4373	1.9968	1.0431
DVGNets-GRU	0.1550	1.4154	1.9546	1.0257
This model	**0.1233**	**0.5829**	**1.5114**	**0.8317**

Table 4
Evaluation results for experiment 3.

Method	IMAPE	IARV	IRMSE	IMDE
DVGNets-UD-GPR-BPNN	0.1306	1.3190	1.7207	0.9156
DVGNets-UD-GPR-SVR	0.1309	0.6483	1.6172	0.8636
DVGNets-UD-GPR-LSTM	0.1351	0.5942	1.9608	1.0810
DVGNets-UD-GPR-GRU	0.1512	0.5410	1.8859	1.0483
DVGNets-UD-GPR-LASSO	0.1471	2.2103	1.7703	0.9442
DVGNets-UD-GPR-CNN	0.1387	2.9021	2.0356	1.0747
DVGNets-UD-GPR-BR	0.1472	2.1140	1.7604	0.9498
DVGNets-UD-GPR-ELM	0.2599	1.3880	2.2642	1.2813
This model	**0.1233**	**0.5829**	**1.5114**	**0.8317**

5. Conclusions

The crude oil price, which is frequently referred to as the backbone of the global industry, holds considerable sway over practitioners, scholars, and policymakers alike. A reliable and consistent forecast of crude oil prices is crucial for economic development, business operations, and investment decisions. However, crude oil prices are characterized by a high degree of chaos and pronounced irregularity, which has rendered certain conventional point prediction methods obsolete. Consequently, an increasing number of scholars are delving into more effective interval prediction techniques.

In the existing link prediction of time series to complex networks, not only the influence of external factors on time series is ignored, but it has

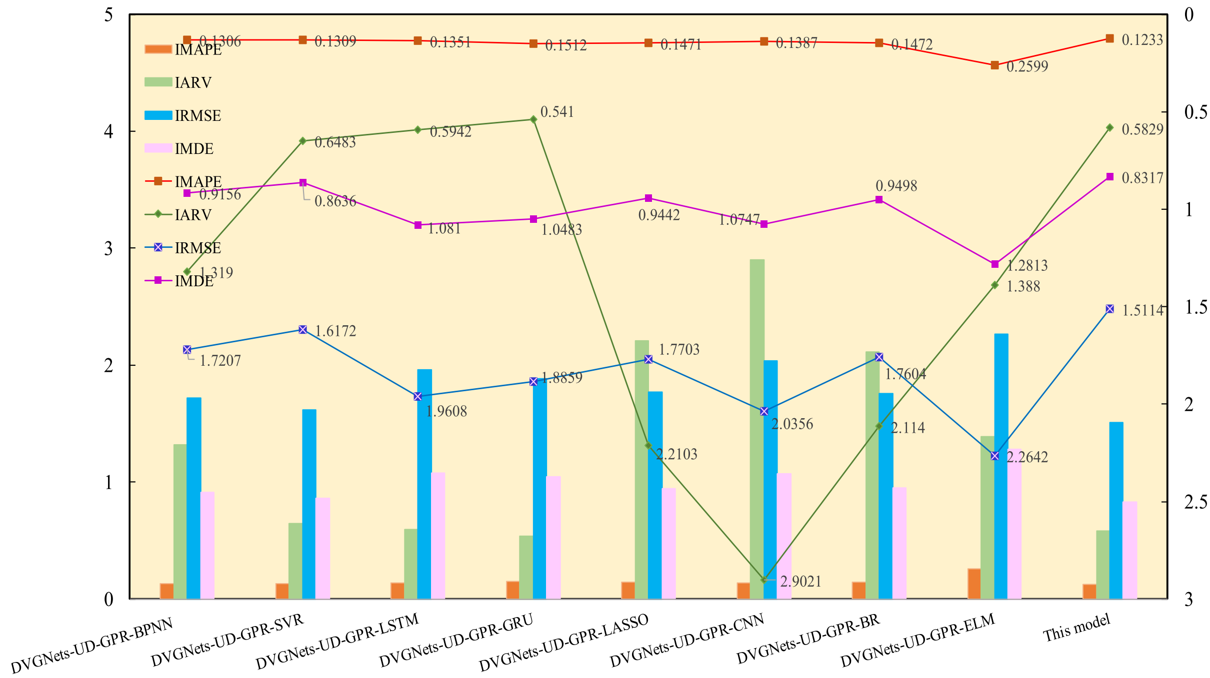

Fig. 13. Comparison of predictive power in experiment 3.

Table 5
The three tests method results between the proposed model and the compared model.

Target model	This model	DM	p-value	$\overline{d}_w$	p-value	$\overline{d}_p$	p-value
Compared model	DVGNets-UD-GPR-SVR	−0.7093	(0.4781)	−0.3309	(0.1618)	−0.0725	(0.2745)
	DVGNets-UD-GPR-BPNN	−1.3696	(0.1708)	−0.6766	(0.2766)	−0.1603	(0.1170)
	DVGNets-UD-GPR-Lasso	−1.9326	(0.0533)	−0.8496	(0.0241)	−0.2074	(0.0310)
	DVGNets-UD-GPR-LSTM	−2.9692	(0.0030)	−1.5604	(0.0047)	−0.4630	(0.0020)
	DVGNets-UD-GPR-CNN	−3.1758	(0.0015)	−1.8594	(0.0021)	−0.4385	(0.0034)
	DVGNets-UD-GPR-ELM	−3.3342	(0.0009)	−2.8422	(0.0044)	−0.6816	(0.0019)
	DVGNets-UD-GPR-GRU	−2.6093	(0.0091)	−1.2722	(0.0076)	−0.4047	(0.0096)
	DVGNets-UD-GPR-BR	−2.1589	(0.0309)	−0.8147	(0.0130)	−0.2149	(0.0190)
	DVGNets-UD-SVR	−0.7123	(0.4763)	−0.4458	(0.3541)	−0.1034	(0.2329)
	DVGNets-UD-LSTM	−1.6070	(0.1081)	−0.9741	(0.1029)	−0.2209	(0.0962)
	DVGNets-UD-BPNN	−2.7253	(0.0064)	−1.5710	(0.0098)	−0.4834	(0.0049)
	DVGNets-UD-GRU	−2.5336	(0.0113)	−1.7029	(0.0044)	−0.4047	(0.0096)
	DVGNets-LSTM	−2.1296	(0.0332)	−1.3900	(0.0381)	−0.3022	(0.0184)
	DVGNets-BPNN	−2.3006	(0.0214)	−1.1397	(0.0147)	−0.2996	(0.0160)
	DVGNets-SVR	−2.4428	(0.0146)	−2.4807	(0.0094)	−0.4764	(0.0083)
	DVGNets-GRU	−2.5380	(0.0111)	−1.5360	(0.0098)	−0.3387	(0.0124)
	SVR	−3.1991	(0.0014)	−1.5440	(0.0023)	−0.7659	(0.0000)
	LSTM	−3.9826	(0.0001)	−1.9496	(0.0001)	−0.5156	(0.0002)
	GRU	−4.4129	(0.0000)	−3.3509	(0.0001)	−0.7659	(0.0000)
	BPNN	−4.4146	(0.0000)	−3.7779	(0.0000)	−0.8091	(0.0001)

never been applied in the field of interval prediction. Therefore, this paper proposes the link prediction model for interval-valued crude oil prices based on complex network and multi-source information. This prediction incorporates the influencing factors of multi-source information into the link prediction of time series to complex networks and is then applied to the prediction problem of interval-valued crude oil prices. To acquire network search indexes associated with crude oil prices, this approach initially employs crawling GRP indicators and international crude oil-related news headlines. At the same time, sentiment analysis is performed on the text to obtain network search data and unstructured data related to sentiment scores; Secondly, using decomposition methods, the center and radius of the interval, as well as their influencing factors, are decomposed and reconstructed into high frequency, low frequency, and trend; Then, with the link prediction concept of DVGNet, the similarity of nodes in the interval subsequence in the directed network is calculated, and the most similar nodes are selected as feature values; Finally, perform a combination prediction and obtain the final interval prediction result by employing the optimal weighted combination.

The experimental findings indicate that the proposed combined prediction method demonstrates robust prediction accuracy and effective generalization capabilities for interval-valued crude oil price time series. Furthermore, the following strengths of the model were identified during the experiment analyses: (1) our approach transforms interval data into a complex network, in contrast to traditional methods for interval-valued crude oil price prediction that disregard the influence of prior interval data. By leveraging link prediction techniques to capture past interval information, we provide a new perspective for interval-valued crude oil price prediction. (2) The impact of multi-source information is not taken into account by the current time series on complex network link prediction methods. We assessed the influence of multi-source information, including the network search index, news title sentiment analysis, and GPR index on crude oil prices, and incorporated them as eigenvalues into link prediction to maximize the accuracy of our predictions. (3) The interval combination model described in this study resolves the uncertainty of prediction method selection, extracts scale features from multiple perspectives, and reduces prediction errors. In addition, the proposed prediction framework is applied to forecast WTI interval-valued crude oil prices, which is in line with the actual demands of the market.

Nevertheless, this study also has certain limitations. Firstly, it does not comprehensively consider multiple sources of data, as it overlooks image and video data related to crude oil prices, which may receive greater attention in real-life scenarios. Secondly, our forecasting model is built on a directed visibility network based on an unweighted static network, focusing solely on the directionality of the network while neglecting the weight and significance of links. Factors considered in computing similarity between nodes are relatively limited. Therefore, future research may focus on analyzing image and video data and developing prediction techniques suitable for weighted dynamic networks.

CRediT authorship contribution statement

Jinpei Liu: Writing – original draft, Methodology, Conceptualization. **Xiaoman Zhao:** Writing – original draft, Data curation. **Rui Luo:** Writing – review & editing, Supervision, Investigation. **Zhifu Tao:** Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgments

The work was supported by the National Natural Science Foundation of China (Nos. 72071001, 72271002, 72001001, 72371001), Outstanding Youth Fund Project for Colleges and Universities of Anhui Province (No. 2023AH020009), Excellent Youth Fund Project for Colleges and Universities of Anhui Province (No. 2023AH030006), Excellent Youth Talent Key Project of Colleges and Universities in Anhui Province (No. gxyqZD2022001).

References

[1] Bai Y, Li X, Yu H, Jia S. Crude oil price forecasting incorporating news text. Int J Forecast 2022;38(1):367–83.
[2] Baumeister C, Kilian L. Forty years of oil price fluctuations: why the price of oil may still surprise us. J Econ Perspect 2016;30(1):139–60.
[3] Maia A, de Carvalho F. Holt's exponential smoothing and neural network models for forecasting interval-valued time series. Int J Forecast 2011;27(3):740–59.
[4] Xiong T, Bao Y, Hu Z, Chiong R. Forecasting interval time series using a fully complex-valued RBF neural network with DPSO and PSO algorithms. Inf Sci 2015; 305:77–92.
[5] Yang W, Han A, Hong Y, Wang S. Analysis of crisis impact on crude oil prices: a new approach with interval time series modeling. Quant Financ 2016;16(12): 1917–28.
[6] Luo R, Liu J, Wang P, Tao Z, Chen H. A multisource data-driven combined forecasting model based on internet search keyword screening method for interval soybean futures price. J Forecast 2024. https://doi.org/10.1002/for.3035.
[7] Mao S, Zeng X. SimVGNets: similarity-based visibility graph networks for carbon price forecasting. Expert Syst Appl 2023;230:120647.
[8] Gao Q, Wen T, Deng Y. A novel network-based and divergence-based time series forecasting method. Inf Sci 2022;612:553–62.
[9] Güne SI, Gündüz-Ö güdücü S, Çataltepe Z. Link prediction using time series of neighborhood-based node similarity scores. Data Min Knowl Disc 2016;30:147–80.
[10] Charfeddine L. Breaks or long range dependence in the energy futures volatility: out-of-sample forecasting and VaR analysis. Econ Model 2016;53:354–74.
[11] Azevedo V, Campos L. Combination of forecasts for the price of crude oil on the spot market. Int J Prod Res 2016;54:5219–35.
[12] Syuhada K, Tjahjono V, Hakim A. Improving value-at-risk forecast using GA-ARMA-GARCH and AI-KDE models. Appl Soft Comput 2023;148:110885.
[13] Abdollahi H, Ebrahimi S. A new hybrid model for forecasting Brent crude oil price. Energy 2020;200:117520.
[14] Wang J, Wang J, Zhang Z, Guo S. Forecasting stock indices with back propagation neural network. Expert Syst Appl 2011;38(11):14346–55.
[15] Li L, Wen S, Tseng M, Chiu A. Photovoltaic array prediction on short-term output power method in centralized power generation system. Ann Oper Res 2020;290: 243–63.
[16] Zhang L, Wang J, Wang B. Energy market prediction with novel long short-term memory network: case study of energy futures index volatility. Energy 2020;211: 118634.
[17] Chiroma H, Abdulkareem S, Herawan T. Evolutionary neural network model for West Texas intermediate crude oil price prediction. Appl Energy 2015;142:266–73.
[18] Sun J, Zhao P, Sun S. A new secondary decomposition-reconstruction-ensemble approach for crude oil price forecasting. Res Policy 2022;77:102762.
[19] Sen A, Choudhury K. Forecasting the crude oil prices for last four decades using deep learning approach. Res Policy 2024;88:104438.
[20] Li J, Hong Z, Zhang C, Wu J, Yu C. A novel hybrid model for crude oil price forecasting based on MEEMD and mix-KELM. Expert Syst Appl 2024;246:123104.
[21] Zheng L, Sun Y, Wang S. A novel interval-based hybrid framework for crude oil price forecasting and trading. Energy Econ 2024;130:107266.
[22] Ginsberg J, Mohebbi M, Patel R, Brammer L, Smolinski M, Brilliant L. Detecting influenza epidemics using search engine query data. Nature 2009;457(7232): 1012–4.
[23] Ravi K, Ravi V. A survey on opinion mining and sentiment analysis: tasks, approaches and applications. Knowl-Based Syst 2015;89:14–46.
[24] Tang L, Zhang C, Li L, Wang S. A multi-scale method for forecasting oil price with multi-factor search engine data. Appl Energy 2020;257:114033.
[25] Yang Y, Guo J, Sun S, Li Y. Forecasting crude oil price with a new hybrid approach and multi-source data. Eng Appl Artif Intell 2021;101:104217.
[26] Wu B, Wang L, Wang S, Zeng Y. Forecasting the US oil markets based on social media information during the COVID-19 pandemic. Energy 2021;226:120403.
[27] Li J, Qian S, Li L, Guo Y, Wu J, Tang L. A novel secondary decomposition method for forecasting crude oil price with twitter sentiment. Energy 2024;290:129954.
[28] Yang K, Cheng Z, Li M, Wang S, Wei Y. Fortify the investment performance of crude oil market by integrating sentiment analysis and an interval-based trading strategy. Appl Energy 2024;353:122102.
[29] Li X, Shang W, Wang S. Text-based crude oil price forecasting: a deep learning approach. Int J Forecast 2019;35(4):1548–60.
[30] Wu B, Wang L, Lv S, Zeng Y. Effective crude oil price forecasting using new text-based and big-data-driven model. Measurement 2021;168:108468.
[31] Mei D, Ma F, Liao Y, Wang L. Geopolitical risk uncertainty and oil future volatility: evidence from MIDAS models. Energy Econ 2020;86:104624.
[32] Ji P, Ye J, Mu Y, Lin W, Tian Y, Hens C, et al. Signal propagation in complex networks. Phys Rep 2023;1017:1–96.
[33] Wang M, Tian L, Zhou P. A novel approach for oil price forecasting based on data fluctuation network. Energy Econ 2018;71:201–12.
[34] Xu H, Wang M, Jiang S, Yang W. Carbon price forecasting with complex network and extreme learning machine. Phys A: Statist Mech Appl 2020;545:122830.
[35] Hu Y, Chu C, Wu P, Hu J. A linear time series analysis of carbon price via a complex network approach. Front Phys 2022;10:1029600.
[36] Yin X, Yang X, Yang Z. Using the R/S method to determine the periodicity of time series. Chaos, Solitons Fractals 2009;39(2):731–45.
[37] Anghinoni L, Zhao L, Ji D, et al. Time series trend detection and forecasting using complex network topology analysis. Neural Netw 2019;117:295–306.
[38] Newman M. Networks: An introduction. Oxford University Press; 2010.
[39] Li X, Li SW. Nonlinear time series forecasting model based on EMD and DVG and its application. Chin J Manage Sci 2022;30(09):275–86.
[40] Lacasa L, Luque B, Ballesteros F, Luque J, Nuno J. From time series to complex networks: the visibility graph. Proc Natl Acad Sci 2008;105(13):4972–5.
[41] Ma Z, Yang X, He A, et al. Complex network recognition of electrocardiograph signals in health and myocardial infarction patients based on multiplex visibility graph. Acta Phys Sin 2022;71(5):48–57.
[42] Yu C, Zhao X, An L, et al. Similarity-based link prediction in social networks: a path and node combined approach. J Inf Sci 2017;43(5):683–95.
[43] Berahmand K, Nasiri E, Forouzandeh S, et al. A preference random walk algorithm for link prediction through mutual influence nodes in complex networks. J King Saud Univ-Comput inform Sci 2022;34(8):5375–87.
[44] Liang X, Wang Y, Yang M. Systemic modeling and prediction of port container throughput using hybrid link analysis in complex networks. Systems 2024;12(1): 23.
[45] Ren W, Jin N, Lei o.. Phase space graph convolutional network for chaotic time series learning. IEEE Trans Industr Inform 2024. https://doi.org/10.1109/TII.2024.3363089.
[46] Jiang M, Chen W, Xu H, et al. A novel interval dual convolutional neural network method for interval-valued stock price prediction. Pattern Recogn 2024;145: 109920.
[47] Wang J, Zhou Y, Jiang H. A novel interval forecasting system based on multi-objective optimization and hybrid data reconstruct strategy. Expert Syst Appl 2023;217:119539.

[48] Wang K, Wang J, Zeng B, et al. An integrated power load point-interval forecasting system based on information entropy and multi-objective optimization. Appl Energy 2022;314:118938.
[49] Quan H, Zhang W, Zhang W, et al. An interval prediction approach of wind power based on skip-GRU and block-bootstrap techniques. IEEE Trans Ind Appl 2023;59 (4):4710–9.
[50] Liu T, Qi S, Qiao X, et al. A hybrid short-term wind power point-interval prediction model based on combination of improved preprocessing methods and entropy weighted GRU quantile regression network. Energy 2024;288:129904.
[51] Sun S, Sun Y, Wang S, et al. Interval decomposition ensemble approach for crude oil price forecasting. Energy Econ 2018;76:274–87.
[52] Wang X, Li X, Li S. Point and interval forecasting system for crude oil price based on complete ensemble extreme-point symmetric mode decomposition with adaptive noise and intelligent optimization algorithm. Appl Energy 2022;328: 120194.
[53] Sun B, Bai J, Chu X, et al. Interval prediction approach to crude oil price based on three-way clustering and decomposition ensemble learning. Appl Soft Comput 2022;123:108933.
[54] Wang P, Tao Z, Liu J, et al. Improving the forecasting accuracy of interval-valued carbon price from a novel multi-scale framework with outliers detection: An improved interval-valued time series analysis mode. Energy Econ 2023;118: 106502.
[55] Wong K, Song H, Witt S, et al. Tourism forecasting: to combine or not to combine? Tour Manag 2007;28(4):1068–78.
[56] Blei D, Ng A, Jordan M. Latent dirichlet allocation. J Mach Learn Res 2003;3: 993–1022.
[57] Boyd-Graber J, Hu Y, Mimno D. Applications of topic models. Foundations and trends®. Inf Retr 2017;11(2–3):143–296.
[58] Abdi H, Williams LJ. Principal component analysis. Wiley Interdiscip Rev: Computat Statist 2010;2(4):433–59.
[59] Liu W, Lü L. Link prediction based on local random walk. Europhys Lett 2010;89 (5):58007.
[60] He Z. Dynamic impacts of crude oil price on Chinese investor sentiment: nonlinear causality and time-varying effect. Int Rev Econ Financ 2020;66:131–53.
[61] Diebold F, Mariano R. Comparing predictive accuracy. J Bus Econ Stat 2002;20(1): 134–44.